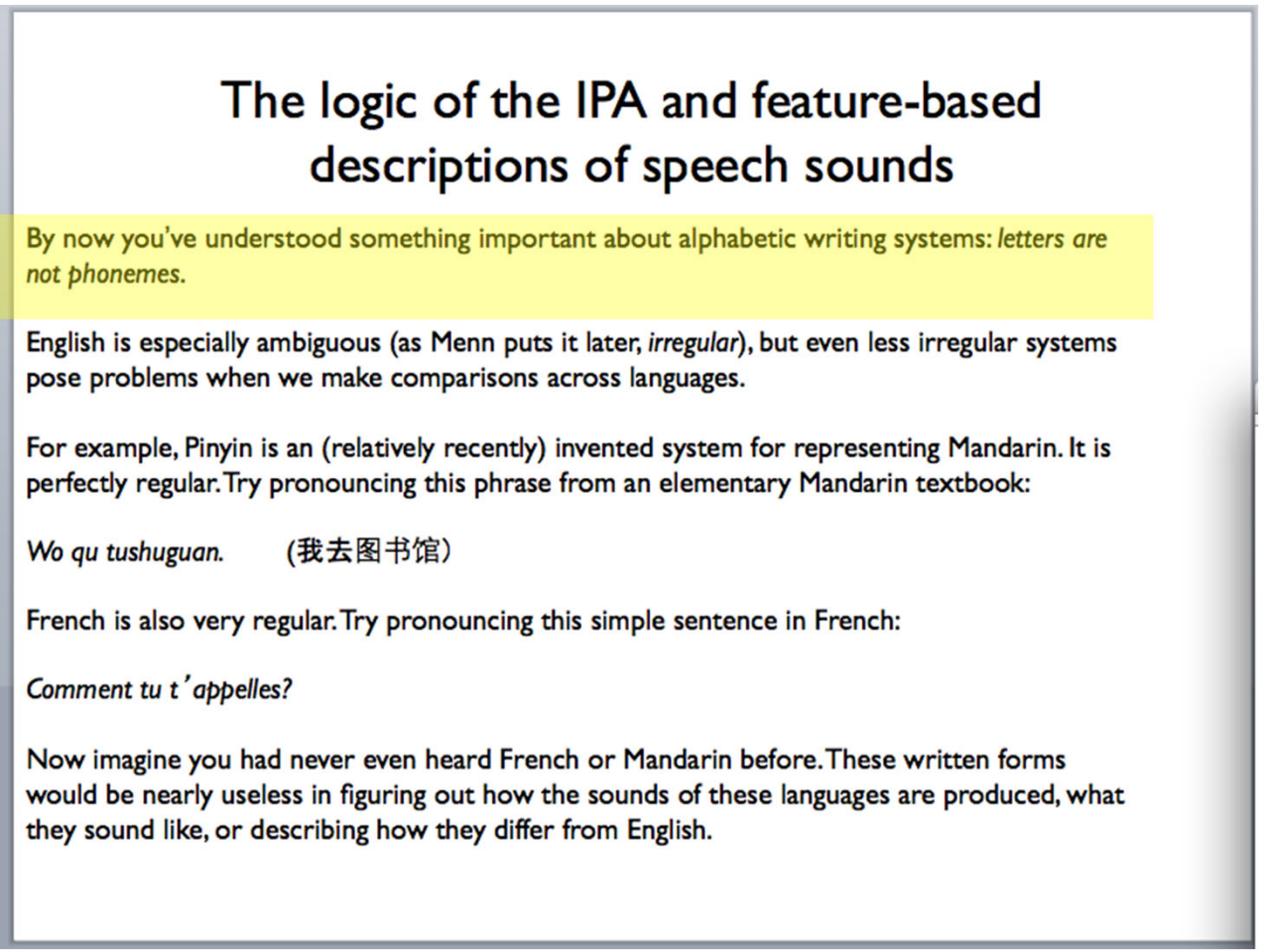
Review Session



The logic of the IPA and feature-based descriptions of speech sounds

By now you've understood something important about alphabetic writing systems: *letters are not phonemes.*

English is especially ambiguous (as Menn puts it later, *irregular*), but even less irregular systems pose problems when we make comparisons across languages.

For example, Pinyin is an (relatively recently) invented system for representing Mandarin. It is perfectly regular. Try pronouncing this phrase from an elementary Mandarin textbook:

Wo qu tushuguan. (我去图书馆)

French is also very regular. Try pronouncing this simple sentence in French:

Comment tu t'appelles?

Now imagine you had never even heard French or Mandarin before. These written forms would be nearly useless in figuring out how the sounds of these languages are produced, what they sound like, or describing how they differ from English.

We were able to demonstrate that phonemes are not letters, but also that *letters are not phonemes* with a "mispronouncing contest" in which people willfully mispronounced words like:

COLONEL

PARADIGM

We also talked about how some scripts (like Pinyin for Mandarin and French) are relatively *regular*.

This difference between scripts is also referred to as *orthographic depth*.

English, due its irregularities is a relatively *deep* script.

Finnish and Korean are very *shallow* scripts.

Listen and repeat:

00:00 00:00



Teija: Juan Ruiz?

Juan: Si! Kyllä, joo! Minä olen Juan Ruiz.

Teija: Hei, minä olen Teija, Teija Virtanen. Tervetuloa Suomeen!

Juan: Kiitos!

http://tasteoffinnish.fi/lesson1-1/

Some writing systems, like Finnish, are very “transparent” (or “shallow” according to the Orthographic Depth Hypothesis).

Let’s listen to this beginning Finnish lesson and follow along with the text…

After about a minute of this, you probably get the idea?

Each letter corresponds more or less unambiguously to one phoneme.

Doubled letters actually correspond to geminated consonants or long vowels.

Finnish children learn to *decode* pronunciations for written forms very very quickly.

Table 3. *Percentage of accurately read pseudowords (means and standard deviations)*

	Grade 1		Grade 2		Grade 3		Grade 4	
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
English	50.3	32.8	71.0	32.5	73.5	28.8	88.2	15.2
French	86.7	5.9	96.7	3.5	98.4	2.4	98.5	2.7
German	88.0	12.4	87.3	9.5	86.0	15.2	87.2	19.5
Dutch	85.2	8.0	88.9	9.1	91.2	8.1	95.1	5.8
Spanish	87.3	12.2	90.4	8.1	90.9	6.8	90.6	8.2
Swedish	93.2	9.6	90.8	10.5	95.4	8.2	97.4	4.3
Finnish	84.9	13.6	89.6	10.1	88.4	8.1	93.7	8.3

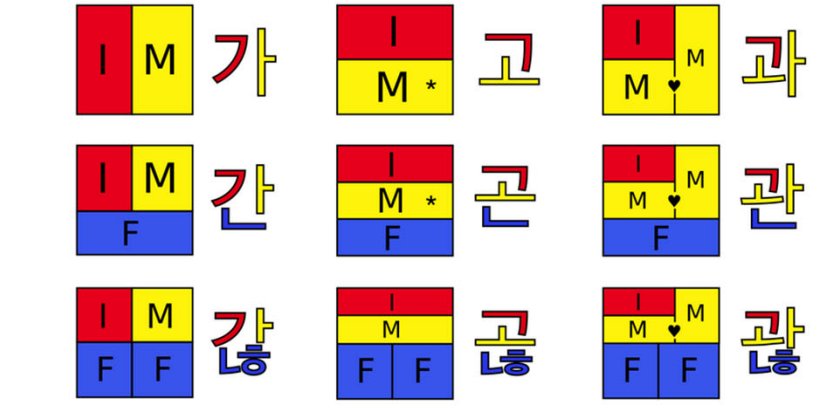
In fact, English is a bit of an outlier among languages that use the Roman alphabet.

One result is that kids learn to decode more slowly in English.

In Grade 1, kids learning a wide variety of other languages are able to generate plausible pronunciations for "pseudowords," (made up words that follow the spelling and sound patterns of their language) at a much higher rate than English learners.

구조 The Placement

I: initial, M: medial, and F: final



Other vowels are used in the 1st placement. * Marked vowels are used in the 2nd placement. ♥ Marked vowels are used in the 3nd placement.

초성 The Initial

ㄱ g ㄲ kk ㅋ k
ㄴ n
ㄷ d ㄸ tt ㅌ t
ㄹ r,l
ㅁ m
ㅂ b ㅃ pp ㅍ p
ㅅ s ㅆ ss
ㅇ No sound
ㅈ j ㅉ jj ㅊ ch
ㅎ h

중성 The Medial

ㅏ a ㅑ ya ♥ㅘ wa
ㅓ eo ㅕ yeo ♥ㅝ wo
* ㅗ o * ㅛ yo
* ㅜ u * ㅠ yu
* ㅡ eu
ㅣ i
ㅐ ae ㅒ yae ♥ㅙ wae
ㅔ e ㅖ ye ♥ㅞ we
♥ㅚ oe
♥ㅟ wi
♥ㅢ ui

Korean is even more regular than Finnish!

The symbols for similar phonemes are represented as similar to one another.

What orthographic feature corresponds roughly to velar place of articulation, for example?

종성 The Final

ㄱ ㄲ ㅋ ㄳ k
ㄴ ㄵ★ㄶ n
ㄷ ㅌ t
ㄹ★ㄺ ㄻ ㄼ★ㄾ l ★ㄽ k ㅀ m ㅄ p
ㅁ m
ㅂ ㅍ ㅄ p
ㅅ ㅆ t
ㅇ ng
ㅈ ㅉ t
ㅎ t

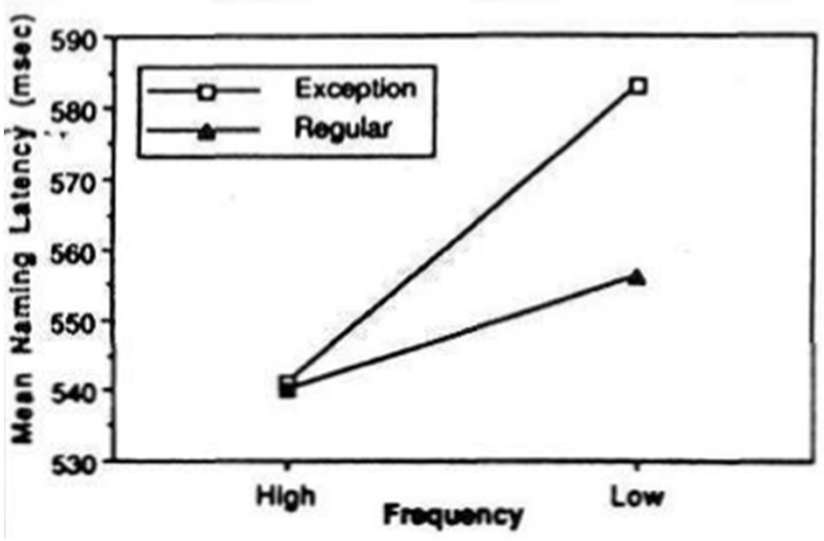
ㄴ(n)/ㄹ(l) +ㄱㄷㅈ -> ㄴ(ŋ)/ㄹ(l)+ㅋㄷㅊ
ㄴ(n)/ㄹ(l) +ㅅ -> ㄴ(n)/ㄹ(l)+ㅆ

ㄺ(k) + ㄱ -> ㄹ(l)

ㄼ(l) of 밟+ㅂ aeum -> ㅂ(p)
ㄼ(l) of 넓 -> ㅂ(p)

Behold: The Frequency by Regularity interaction.

The frequency effect is much larger for exception words than regular words.



This is a landmark finding in psycholinguistics.

We'll see how it has impacted a number of other areas (most immediately inflectional morphology).

What does it mean?

One interpretation

There are two separate processes in word recognition:

1. A sublexical process (decoding), and

2. A lexical process (knowledge of a particular word)

For high frequency (HF) words, the lexical process is so fast and efficient that there is little influence of the sublexical process.

The sublexical process only shows up when we read nonwords, or low frequency (LF) words.

PDP models of visual word recognition

Seidenberg & McClelland (1989)

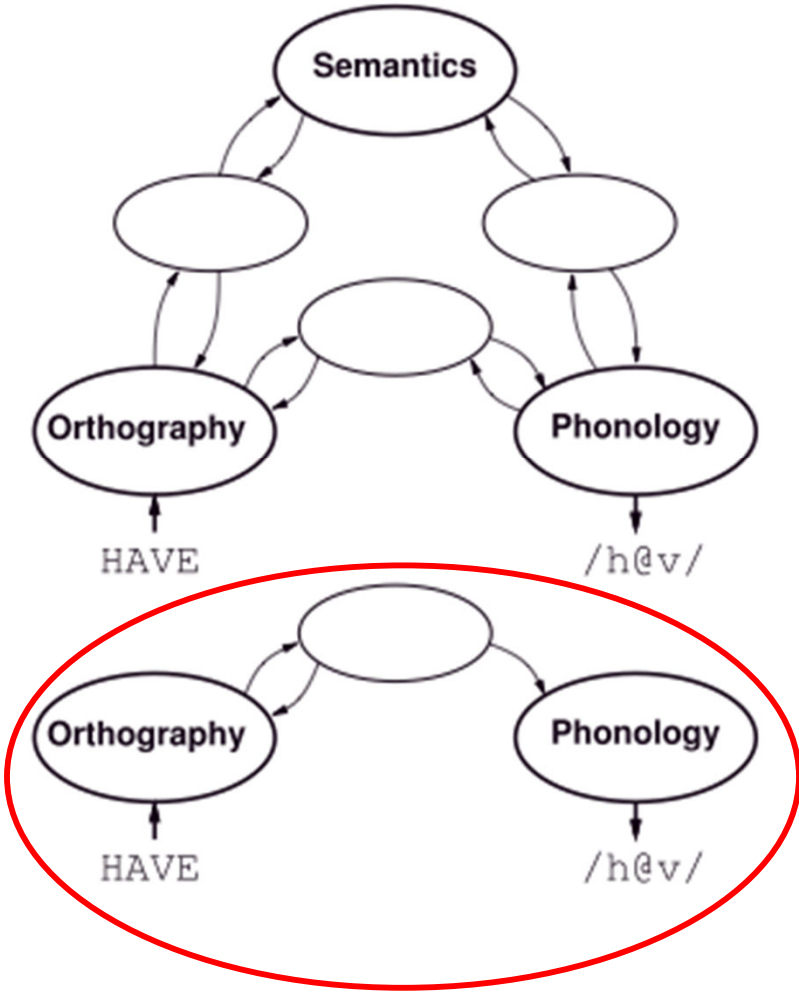
Different assumptions from dual-route models:

- "rules" are just generalizations from lots of words, not a separate process

- the "lexicon" is also distributed. Some whole-word information is stored in O → P mappings, some in other parts of the system.

Different research strategy from dual-route models:

Explain reading in terms of domain-general *learning* mechanisms, rather than reverse-engineer the system from adult/neuropsych data.

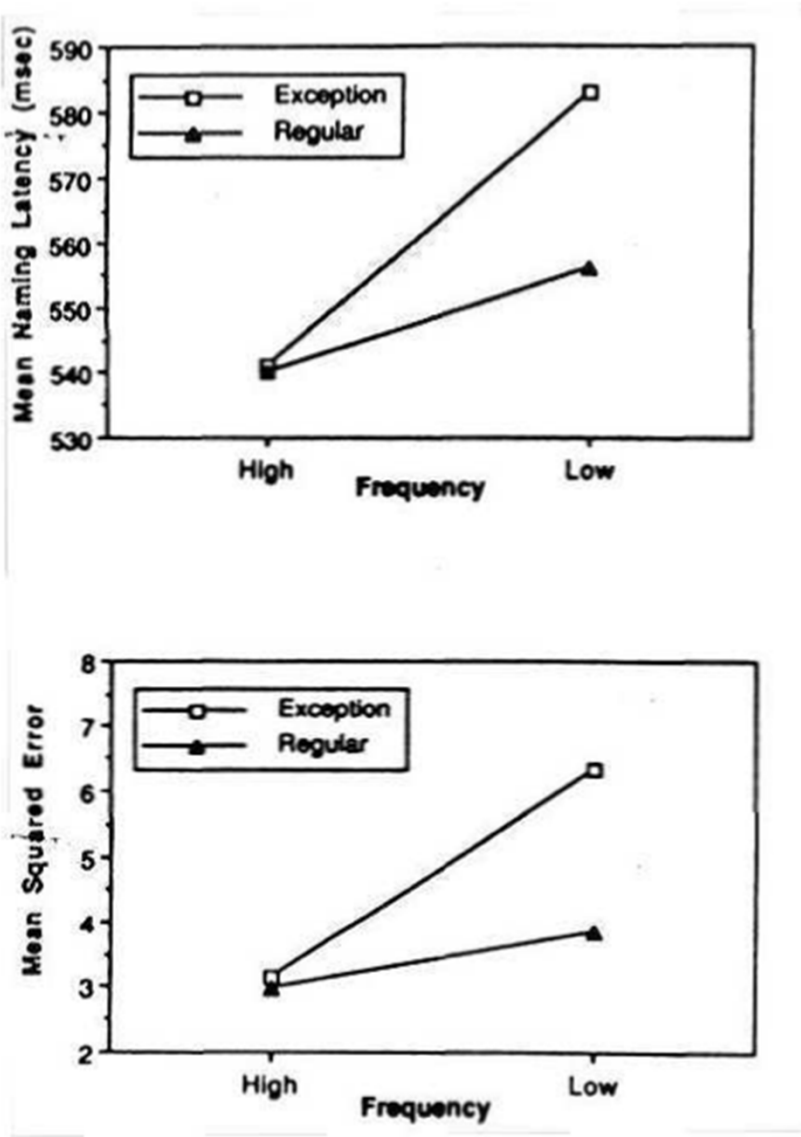


Frequency by regularity interaction

The model captured the interaction between frequency and regularity without assuming that these effects resulted from different processes.

How?

- "regular" words are *consistent* with statistical patterns from many other words.
- "exception" words are *inconsistent* with these statistics.
- frequency also impacts how well words are learned. So, highly frequent words are less effected by inconsistency with other words.



Reading words and nonwords in the same "route"

In principle:

- nonwords named via the same set of weights used for words.

- generalization takes advantage of the same statistical patterns that give rise to regularity effects.

In practice, though, not so great…

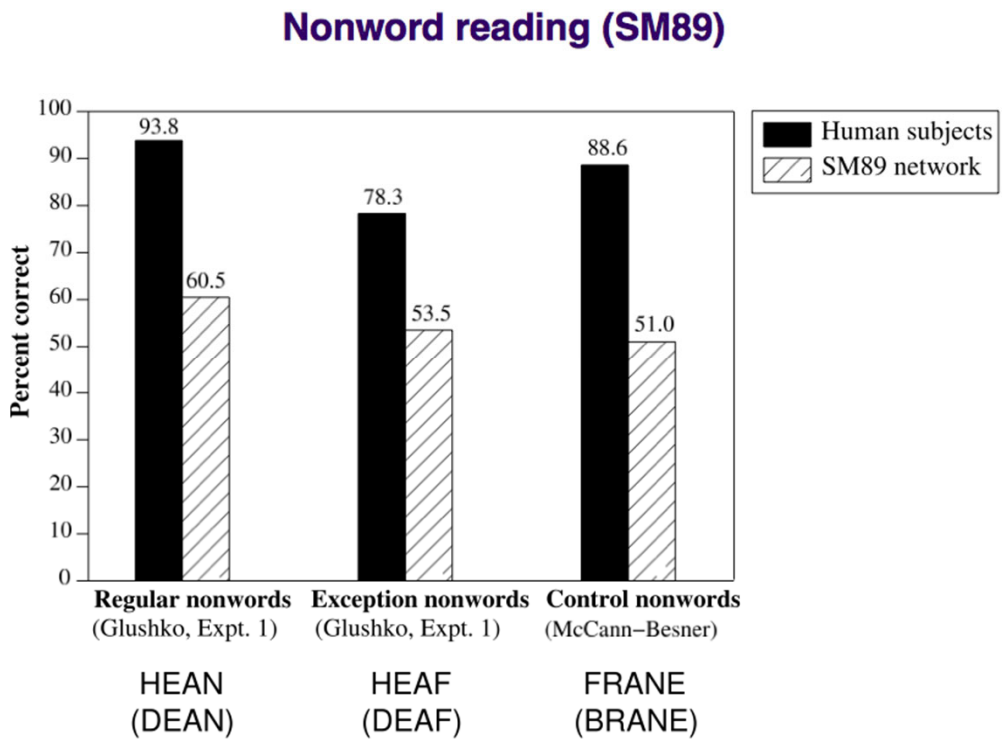
Nonword accuracy was not very good.

Why?

This is a classic case of ambiguity in model evaluation.

Was the theoretical claim -- that nonwords and exceptions could be read by the same, associative mechanism – wrong?

Or, was the problem something specific to the way the model was made?

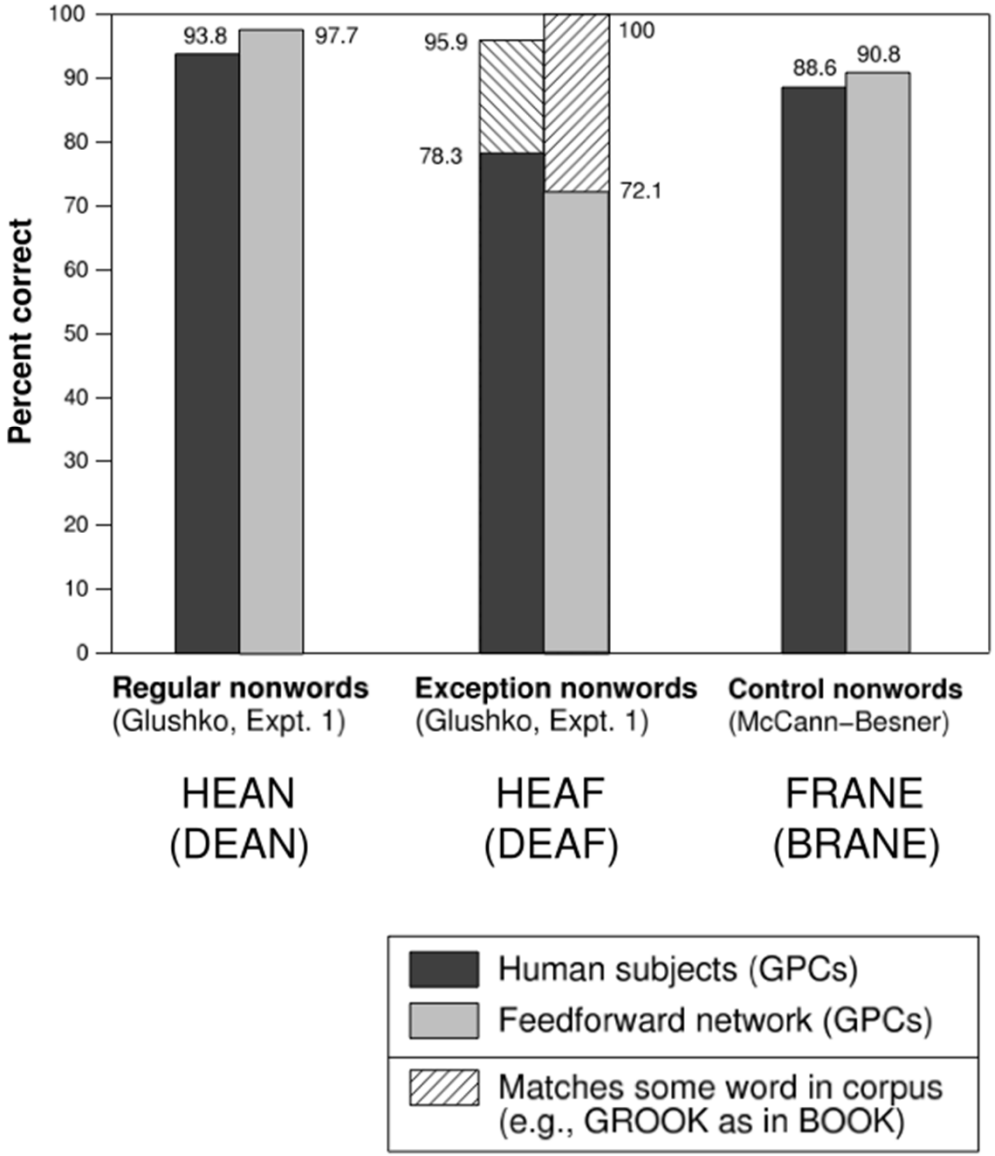


Nonword reading related to phonological representations

Many studies demonstrate a relationship between "decoding" (i.e., nonword reading) and phonological processing.

The phonological representations used in Seidenberg & McClelland were sort of weird: Wickelfeatures.

Indeed, a model by Plaut et al. (1996, Psych Review) improved the phonological representations, and produced better performance on nonwords.



How are phonological representations and nonword generalization related?

Generalization depends on similarity. (In fact this principle is inherited from behaviorism.)

Poor phonological representations (like Wickelfeatures) fail to adequately capture the similarity space of words in the language.

Word-specific information is more robust to poor representations, because it doesn’t depend entirely on generalization.

How are phonological representations and nonword generalization related?

Harm & Seidenberg (1999) investigated this more systematically, by manipulating the amount of "noise" in the phonological representations.

They were able to replicate a number of key findings in the literature on developmental dyslexia:

- relationship between *degree* of phonological deficit and *phonological dyslexia*

- relative sparing of word recognition

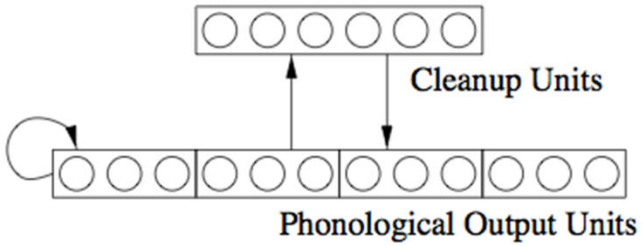


Figure 1. Architecture of the phonological attractor network.

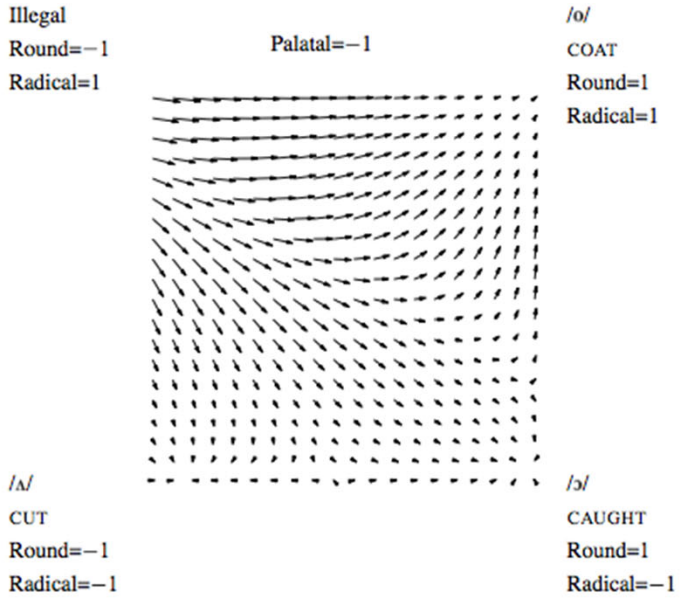
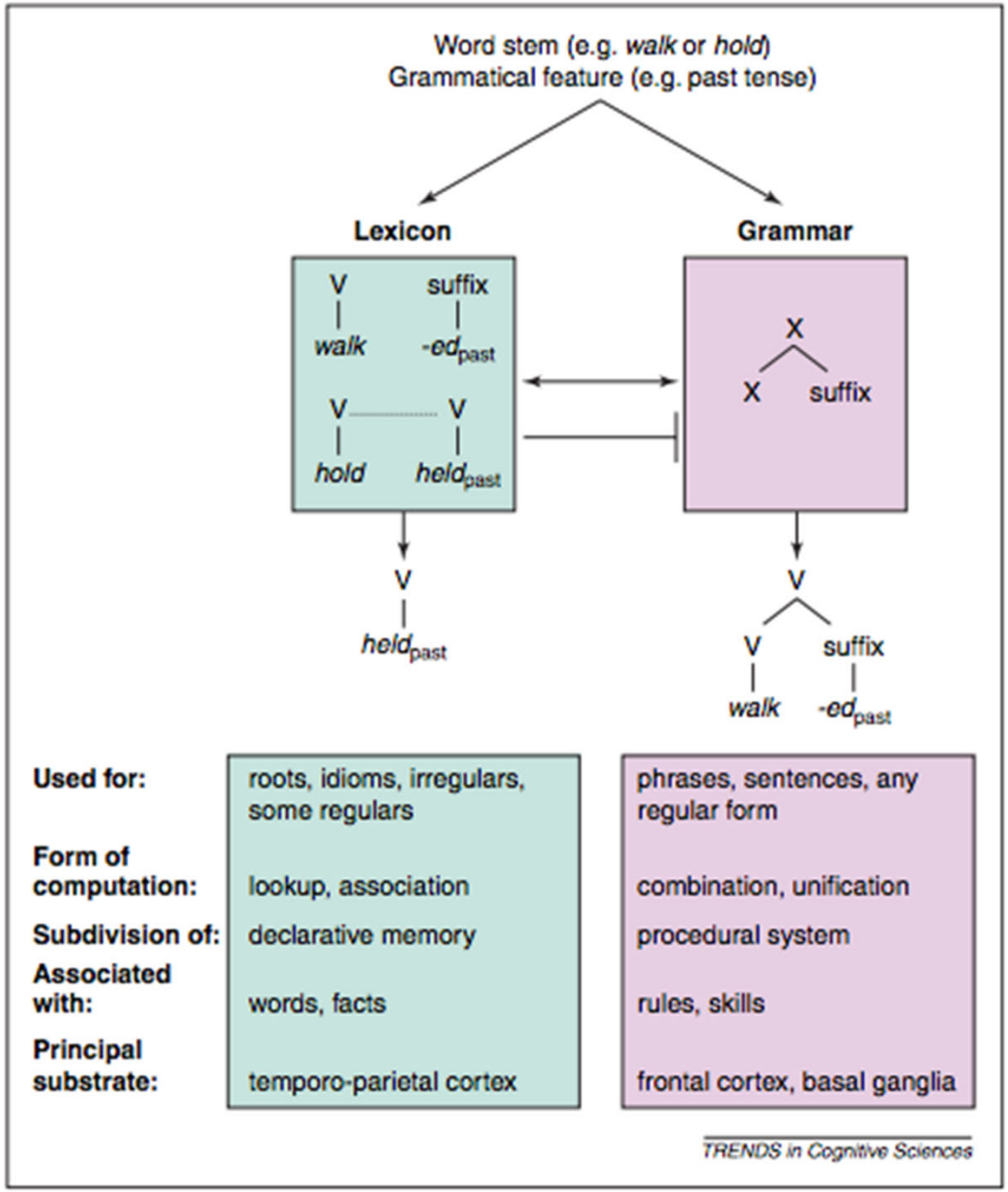


Figure 6. Phonological attractor, depicting 3 legal phonemes derived from combinations of two features: round and radical.

A “dual route” model for the past tense

How is this similar to the dual-route model we were just considering for reading?

How is it different?



Pinker & Ullman (2002, TICS)

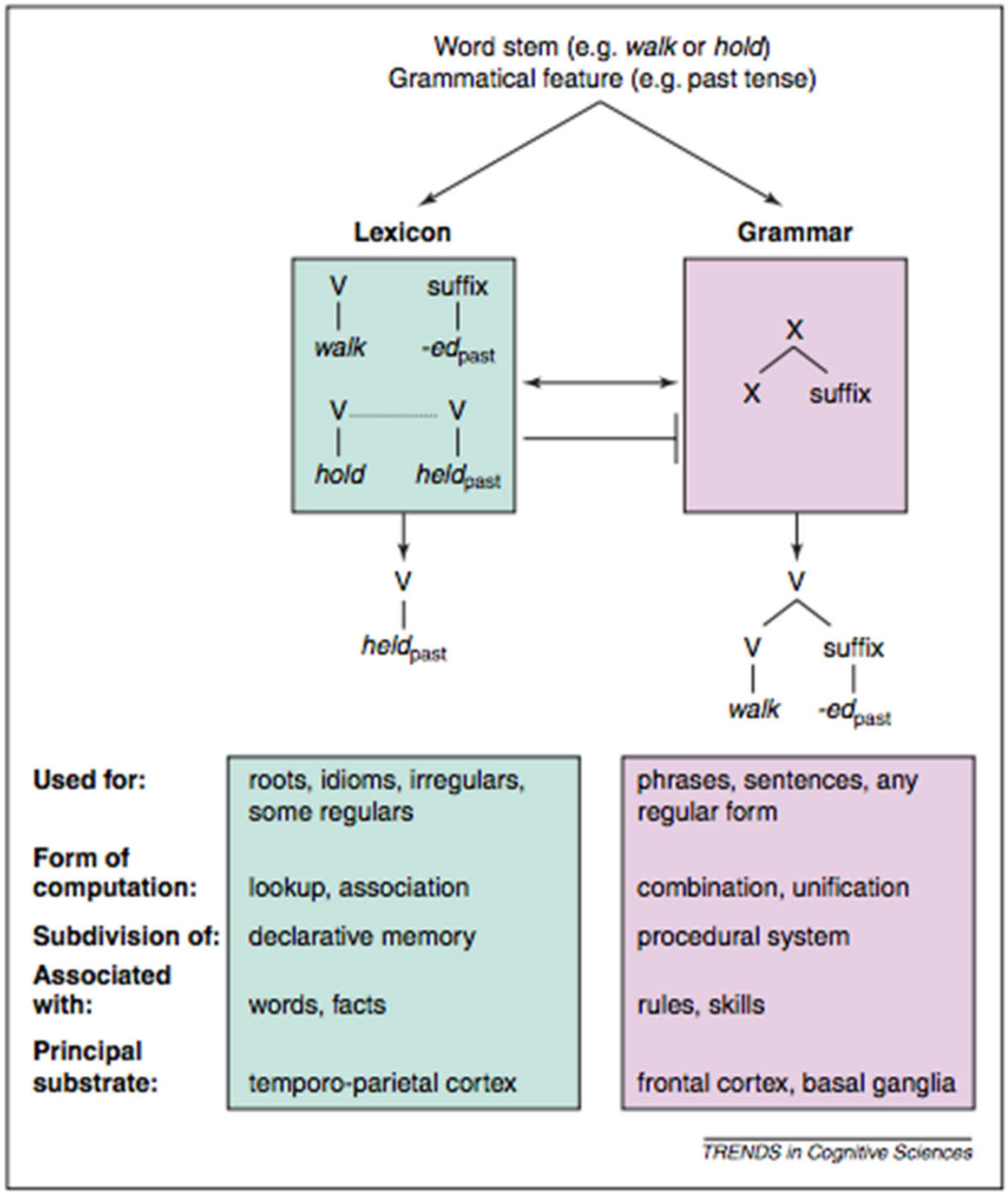
Similar architecture, different origins

Remember the dual route model was developed based on neuropsychological double-dissociations.

Pinker et al. are motivated by other concerns.

Generative linguistics of the time was mostly about categorical, symbolic, rule-based grammars.

These rules often have exceptions.



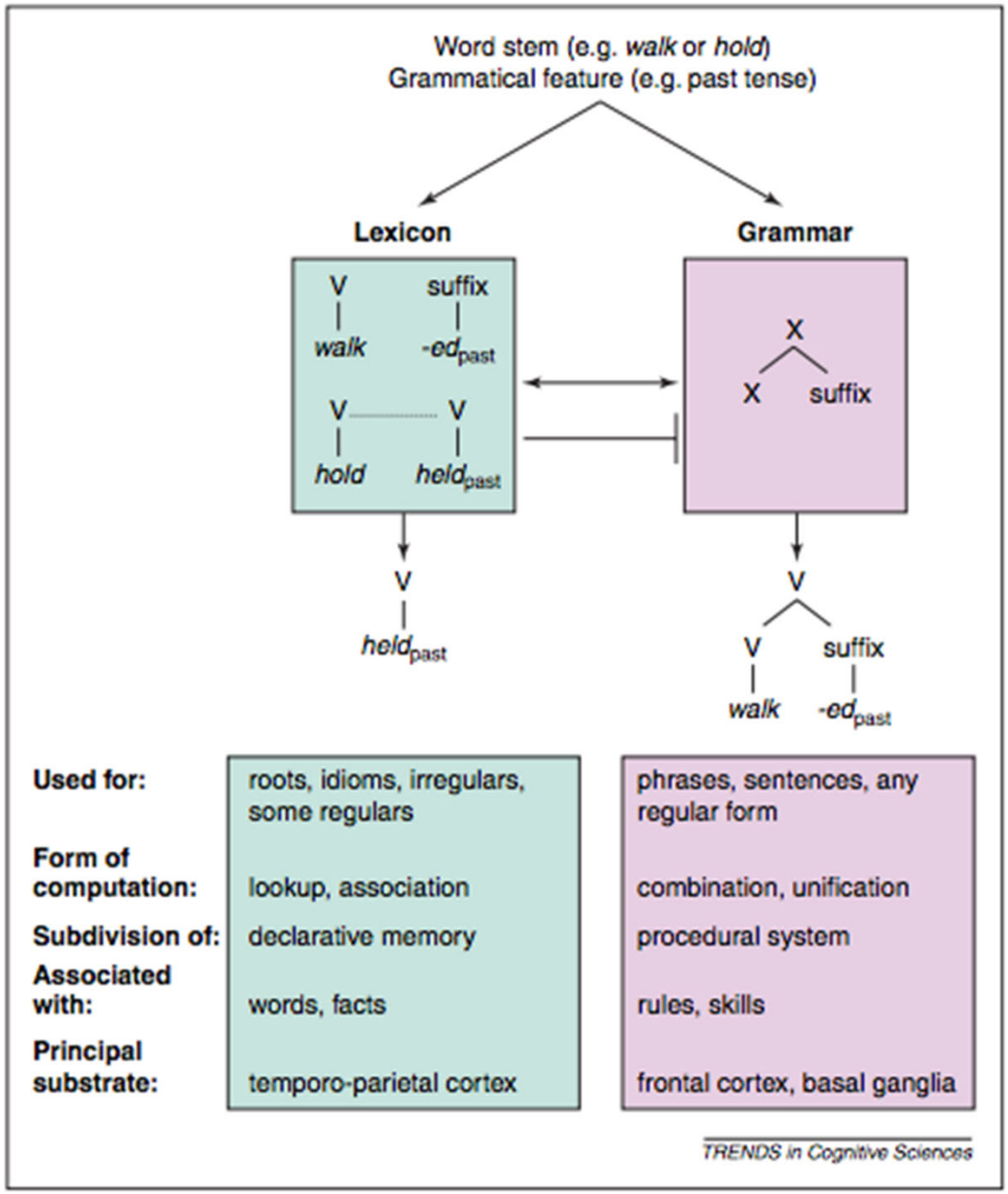
Similar architecture, different origins

Rather than make the grammar inelegant by adding machinery to handle the exceptions, linguists would often just leave the exceptions out of the grammar.

So dual-process model proposes a way to "handle" the exceptions while keeping the grammar elegant, symbolic, categorical, etc.

This second process had all the properties the rule route did not. It was:

- word-specific (lexical)
- sensitive to frequency of occurrence
- sensitive to phonological similarity (keep-kept, weep-wept, leap-leapt, etc.)



An alternative approach: It's all associative learning!

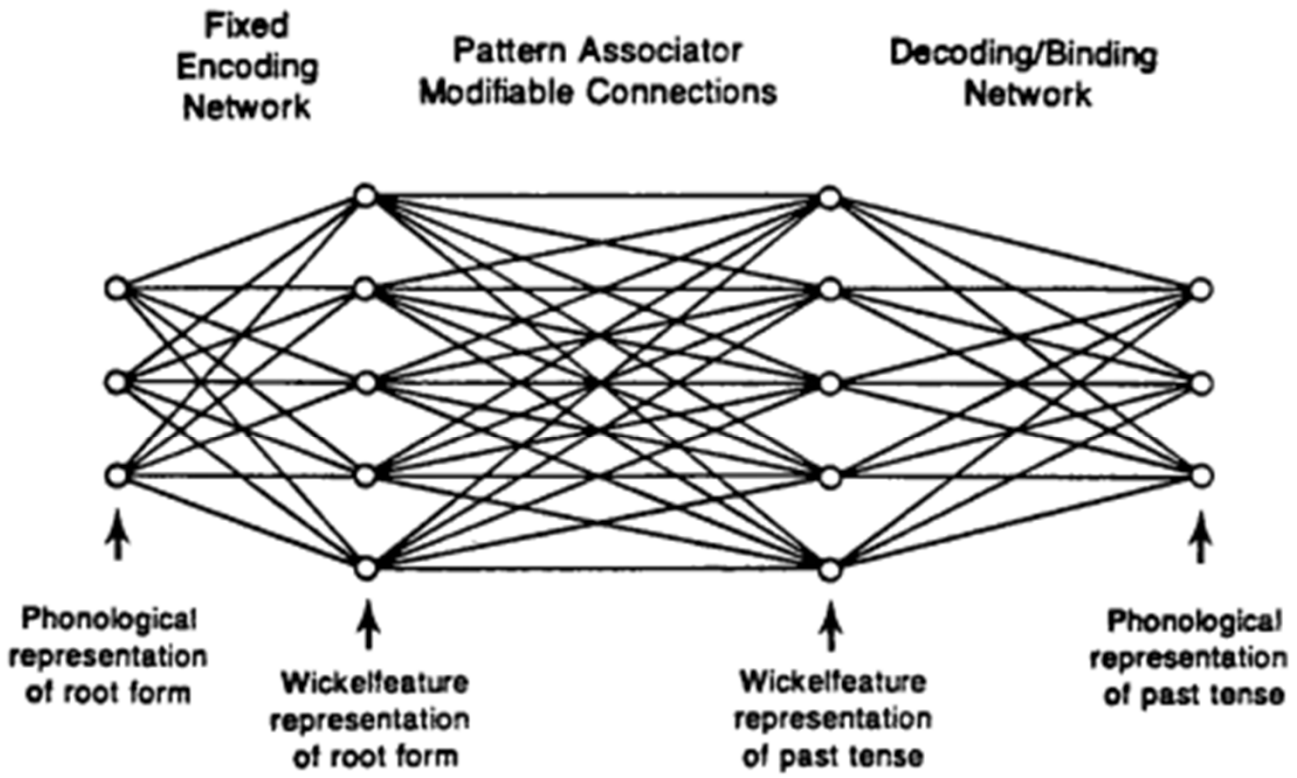


FIGURE 1. The basic structure of the model.

Rumelhart & McClelland (1987) available, along with lots of these other papers, here: http://psych.stanford.edu/~jlm/papers/

Target phenomena of Rumelhart & McClelland

Because this was one of the first *learning models* to address a language process, the focus was on capturing developmental phenomena.

Specifically: “U-shaped” learning:

- some frequent irregular verbs are learned quite early, e.g., *run-ran*

- then, spontaneously, children begin producing “regularizations:” *runned.*

- eventually, they start producing correct forms again.

Learning is “U-shaped” because children often get worse before they get better, so performance goes down before it goes back up.

Why is U-shaped learning important to the "words and rules" approach?

It's understood in terms of distinct stages/processes:

- early correct forms are just imitations, don't depend on the same mechanisms as adult processing.

- when the rule is initially learned, it is applied to all cases. We can think of the rule as the "best hypothesis" for all items generally.

- over time, exceptions are learned associatively, one-by-one and incorporated into lexical knowledge. Further, the "blocking process" develops.

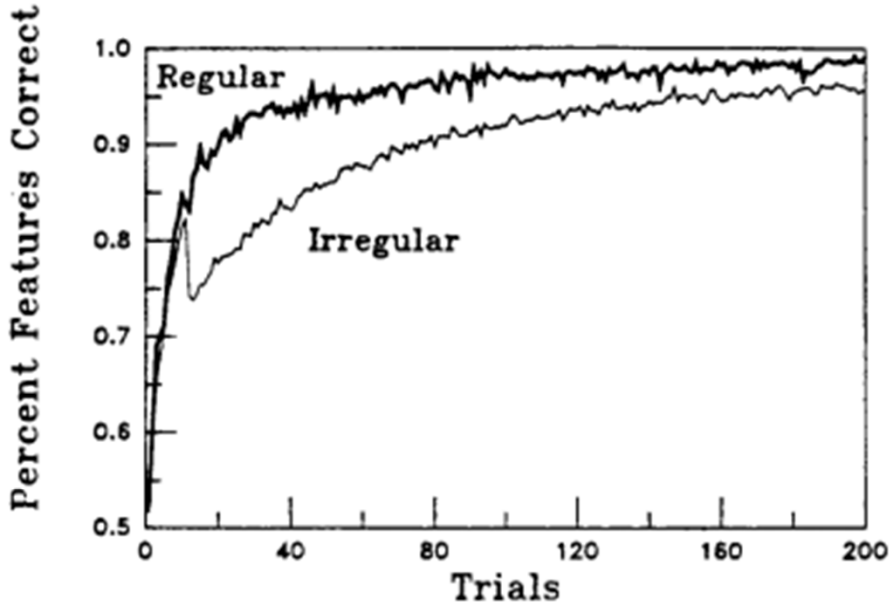


FIGURE 4. The percentage of correct features for regular and irregular high-frequency verbs as a function of trials.

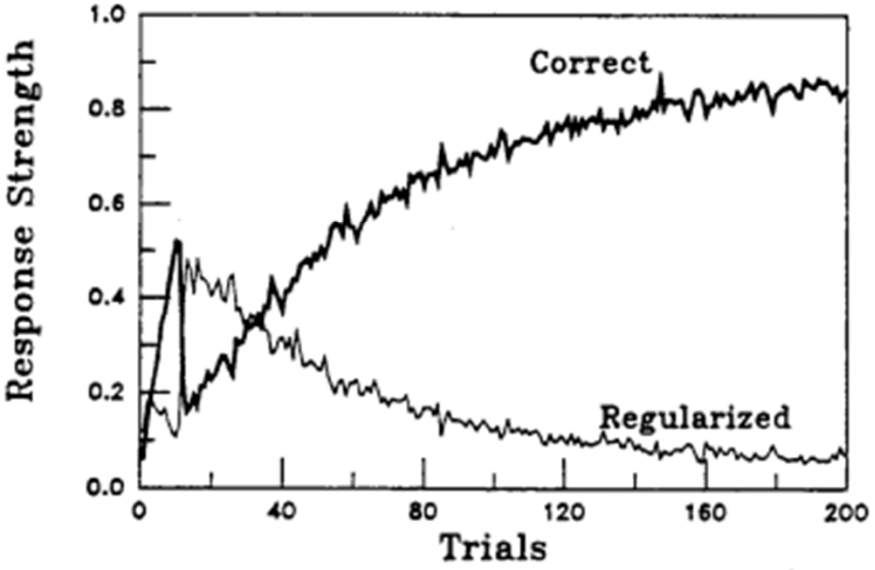


FIGURE 5. Response strengths for the high-frequency irregular verbs. The response strengths for the correct responses are compared with those for the regularized alternatives as a function of trials.

"U-shaped learning" in Rumelhart & McClellands's model.

- training set is based on data about input to children, which changes over time.

- Early on, irregular verbs predominate – in fact the highest-frequency verbs tend to have irregular past-tense forms: GIVE, TAKE, HAVE, GO, etc.

- As the vocabulary expands to include more regular forms, the model reorganizes to capture those regularities.

-- this interferes with some of the early-learned forms.

TABLE 18

SYSTEM RESPONSES TO UNFAMILIAR LOW-FREQUENCY REGULAR VERBS

Verb Type	Presented Word	Phonetic Input	Phonetic Response	English Rendition	Response Rendition
End in *t/d*	guard	/gɑrd/	/gɑrd/	(guard)	0.29
			/gɑrd˜d/	(guarded)	0.26
	kid	/kid/	/kid/	(kid)	0.39
			/kid˜d/	(kidded)	0.24
	mate	/mAt/	/mAt˜d/	(mated)	0.43
			/mAd˜d/	(maded)	0.23
	squat	/skw*t/	/skw*t˜d/	(squated)	0.27
			/skw*t/	(squat)	0.22
			/skw*kt/	(squawked)	0.21
End in unvoiced consonant	carp	/kɑrp/	/kɑrpt/	(carped)	0.28
			/kɑpt˜d/	(carpted)	0.21
	drip	/drip/	/dript˜d/	(dripted)	0.28
			/dript/	(dripped)	0.22
	map	/map/	/mapt˜d/	(mapted)	0.24
			/mapt/	(mapped)	0.22
	shape	/sAp/	/sApt/	(shaped)	0.43
			/sipt/	(shipped)	0.27
	sip	/sip/	/sipt/	(sipped)	0.42
			/sept/	(sepped)	0.28
	slip	/slip/	/slept/	(slept)	0.40
	smoke	/smOk/	/smOkt˜d/	(smokted)	0.29
			/smOk/	(smoke)	0.22
	snap	/snap/	/snapt˜d/	(snapted)	0.40
	step	/step/	/stept˜d/	(stepted)	0.59
	type	/tip/	/tipt˜d/	(typted)	0.33
End in voiced consonant or vowel	brown	/brwn/	/brwnd/	(browned)	0.46
			/br*nd/	(brawned)	0.39
	hug	/h˜g/	/h˜g/	(hug)	0.59
	mail	/mA˜l/	/mA˜ld/	(mailed)	0.38
			/memb˜ld/	(membled)	0.23
	tour	/tur/	/turd˜r/	(toureder)	0.31
			/turd/	(toured)	0.25

Generalization testing:

The model was trained on a small number of verbs, so they could do a “wug” test with real words…

TABLE 16

PROPORTION OF WICKELFEATURES CORRECTLY GENERATED

Regular	.92
Irregular	.84
Overall	.91

But…

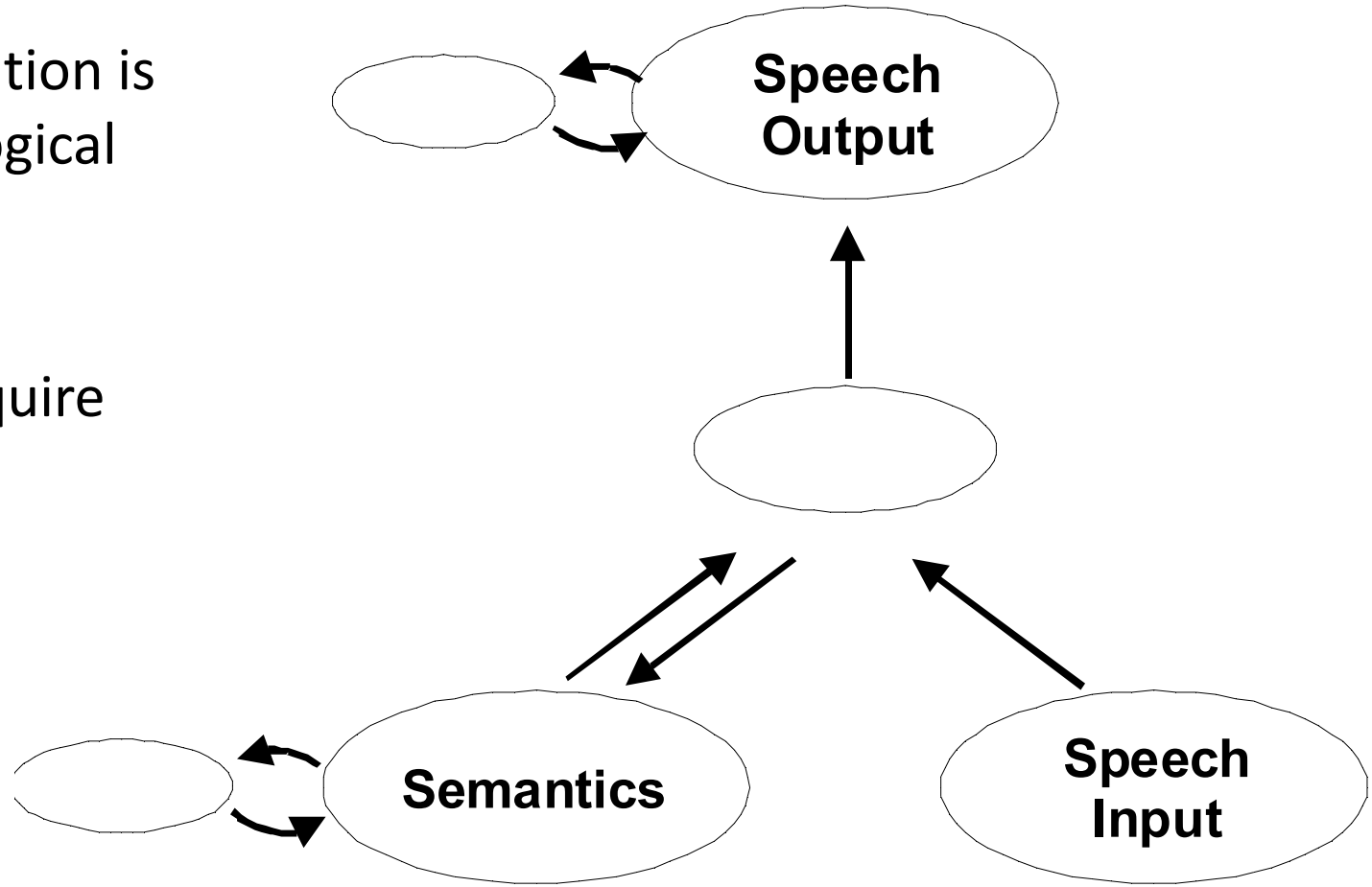
Finally, there were just four responses that involved the addition or modification of consonants. These were *maded* as a past tense of *mate*, *squawked* as a past tense for *squat, membled* as a past tense for *mail*, and *toureder* as a past tense for *tour*. It is unlikely that humans would make these errors, especially the last two, but these responses are, for the most part, near threshold. Furthermore, it seems likely that many of these responses could be filtered out if the model incorporated an auto-associative network of connections among the output units. Such a network could be used to clean up the output pattern and would probably increase the tendency of the model to avoid bizarre responses. Unfortunately, we have not yet had the chance to implement this suggestion.

A new(er) model: Joanisse & Seidenberg (1999)

Theoretical premise:

- "rule-like" generalization is supported by phonological representations

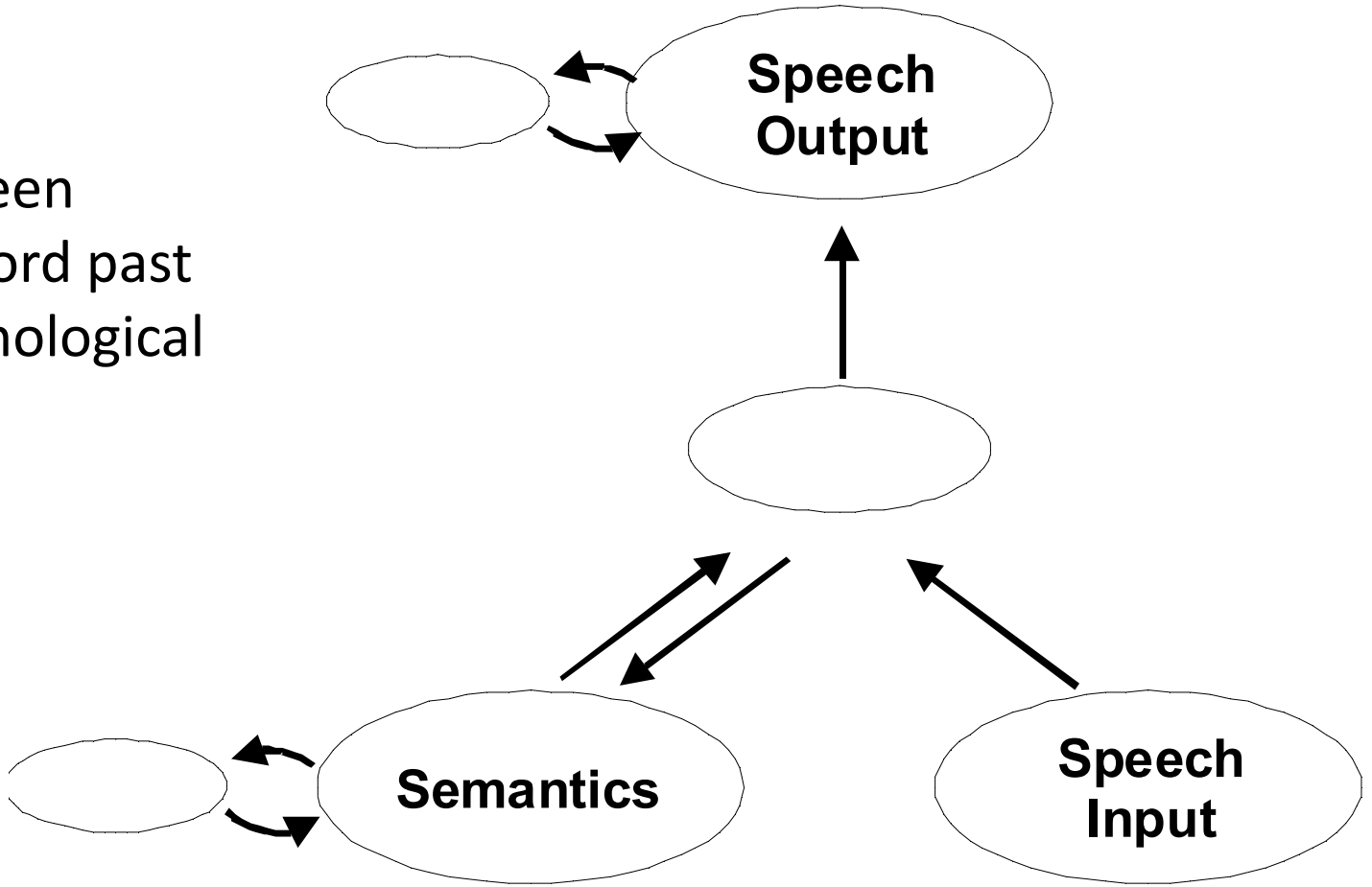
- "irregular" forms require greater support from semantics



A new(er) model: Joanisse & Seidenberg (1999)

Target phenomenon:

- dissociation between irregular and nonword past tense in neuropsychological data.



Neuropsychological data

Broca's Aphasia

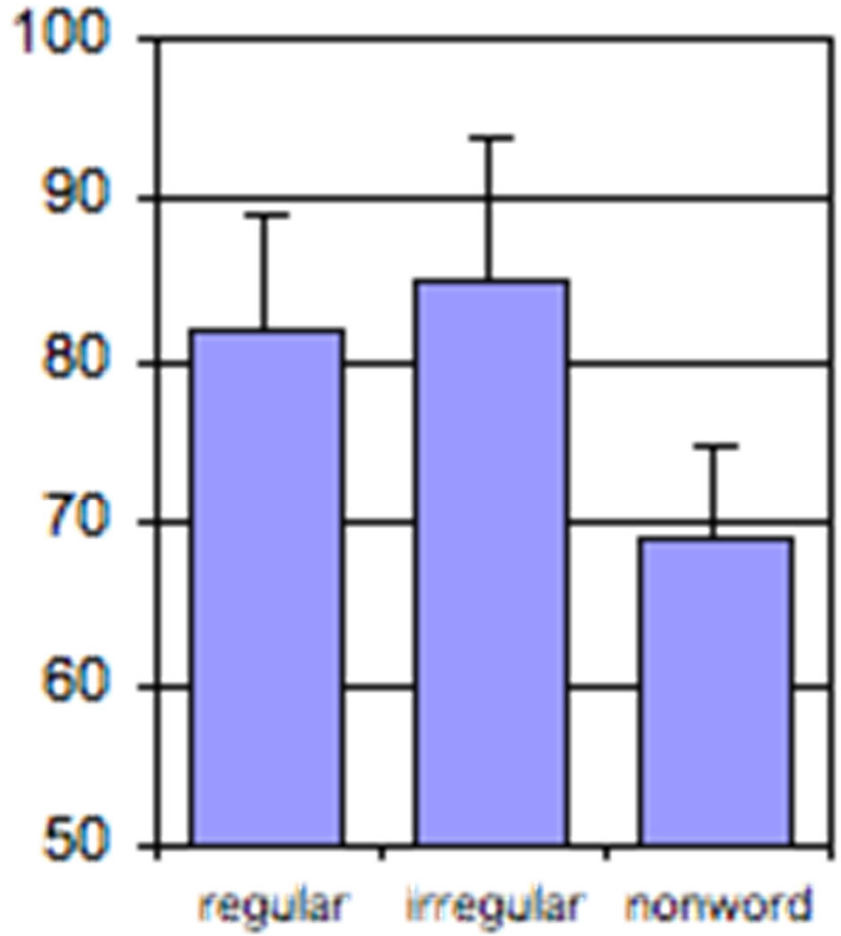
Injury to left anterior brain regions

Poor speech production

Problems with regular morphology, esp. nonwords

wug → *wugged,*

blick → *blicked*

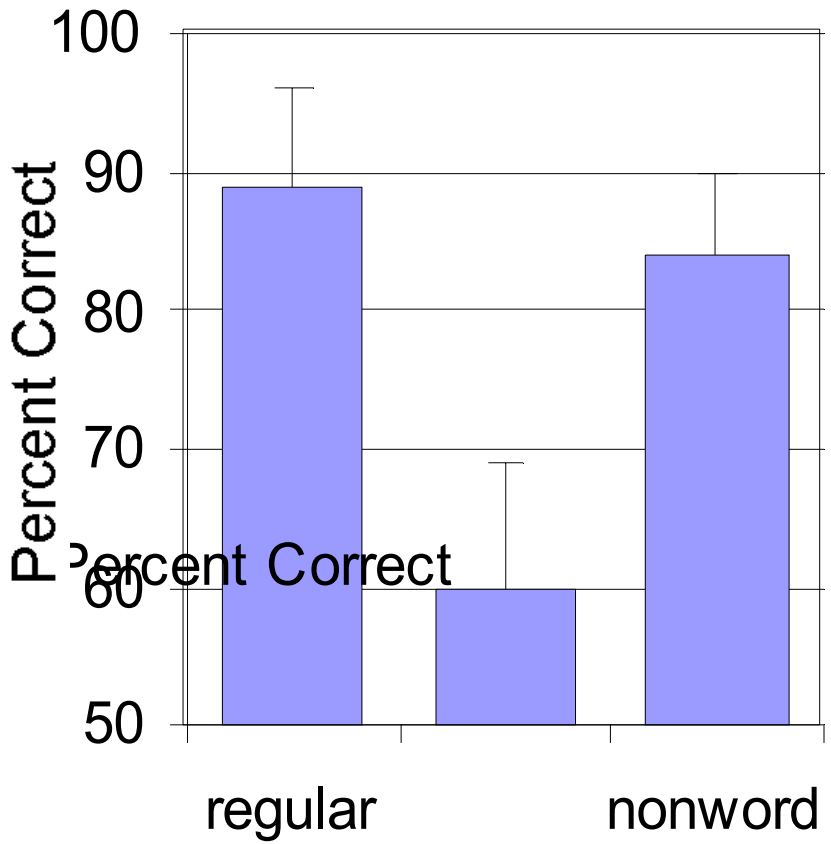


Neuropsychological data: Problems with irregulars

Wernicke's and Alzheimer's patients

Anomic patients with injury to inferior/middle temporal brain regions

Poor on irregulars (middle bar)



An attractive dichotomy:

Wernicke's aphasia = "Word salad" or "Jabberwocky," often with neologisms. Syntax roughly intact:

> One Wernicke's aphasia patient, for example, replied to a question about his health with: "I felt worse because I can no longer keep in mind from the mind of the minds to keep me from mind and up to the ear which can be to find among ourselves" (Kriendler et al., 1971). Another patient, when asked about his poor vision said: "My wires don't hire right". And an aphasic physician, asked if he was a doctor, replied: "Me? Yes, sir, I'm, a male demaploze on my own. I still know my tubaboys what for I have that's gone hell and some of them go". One aphasic described a fork as "a need for a schedule" and a spoon as "How many schemes on your throat".

Broca's aphasia = "Telegraphic" speech, difficulty with word order and grammatical particles (relative sparing of important content words):

> These fluent but uninterpretable utterances are very different from the Broca aphasic's answer when asked what brought him to the hospital: "Yes – ah – Monday – ah – Dad and P.H. (the patient's name) and Dad ... hospital. Two ... ah doctors ..., and ah ... thirty minutes ... and yes ... ah ... hospital. And, er Wednesday ... nine o'clock and eh Thursday, ten o'clock ... doctors. Two doctors ... and ah ... teeth. Yeah ... fine" (Badecker and Caramazza, 1985).

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Note that this is fairly distant from how Broca described his original cases.

Remember this slide from a few weeks ago?

This is confusing: *Broca's actual patients would not necessarily be classified as Broca's aphasics.* They were arguably unable to produce speech at all, apparently due to motor control issues, and would be classified as suffering from *apraxia of speech.*

Paul Broca's presentation of Leborgne (or "Tan")



Behavioral description:

- could only produce one jargon syllable
- seemed otherwise to be in possession of his senses
- in fact, seemed to *understand what was being said.*

Consider:

- the role of dissociations in this argument: why is it critical that Leborgne has other senses intact?

- why does a "capacity for articulate speech" sound like a more plausible function of a particular region of the brain than "criminality" or "philopropogenitiveness?"

- if you could examine Tan, what measurements would you want to make in order to specify his deficits in terms of the levels of description used in modern linguistics (e.g., phonetic, phonological, lexical, syntactic)?

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

This dichotomy is interesting in part because it seems well aligned with a sharp distinction drawn late 20th Century formal linguistics between the *grammar* and the *lexicon*.

This starts with Chomsky's formulation of the computational problem of language learning, and compelling demonstrations of the independence of syntax and semantics.

It appears that we recognize a new item as a sentence not because it matches some familiar item in any simple way, but because it is generated by the grammar that each individual has somehow and in some form internalized. And we understand a new sentence, in part, because we are somehow capable of determining the process by which this sentence is derived in this grammar.

"Neither (a) 'colorless green ideas sleep furiously' nor (b) 'furiously sleep ideas green colorless', nor any of their parts, has ever occurred in the past linguistic experience of an English speaker. But (a) is grammatical, while (b) is not." (Chomsky, 1955)

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Consider, too, the difference between the Functional and Positional levels in our speech production model.

Lemmas have an *arbitrary* relationship to their referents.

But the **syntactic trees**, **affixes**, etc. at the positional level are the product of *abstract syntactic processes.*

"But," you might object "what about words like HOUSEBOAT and DOGHOUSE, where the meaning is clearly not arbitrary," and I would remind you of speech error evidence that *derivational morphology* is done at the positional level, e.g., *easy enoughly.*

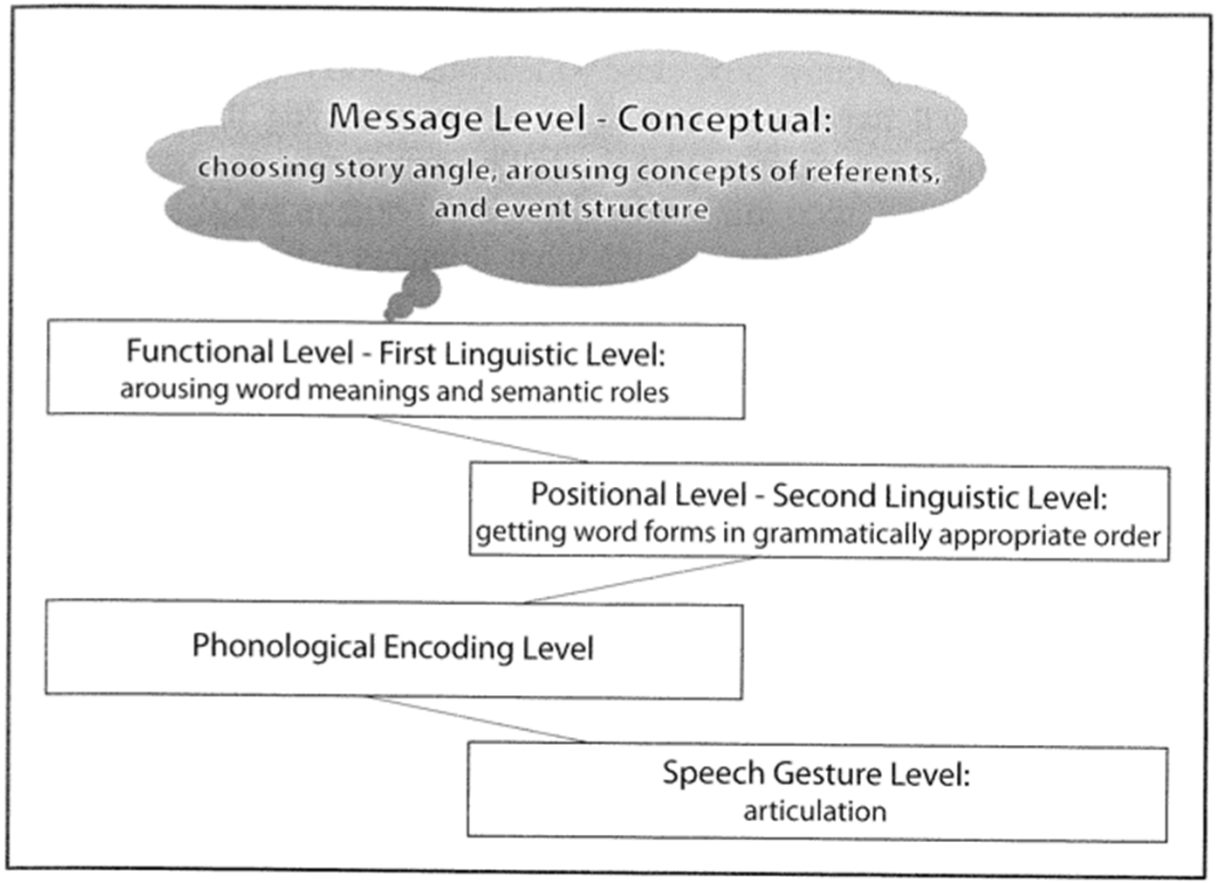
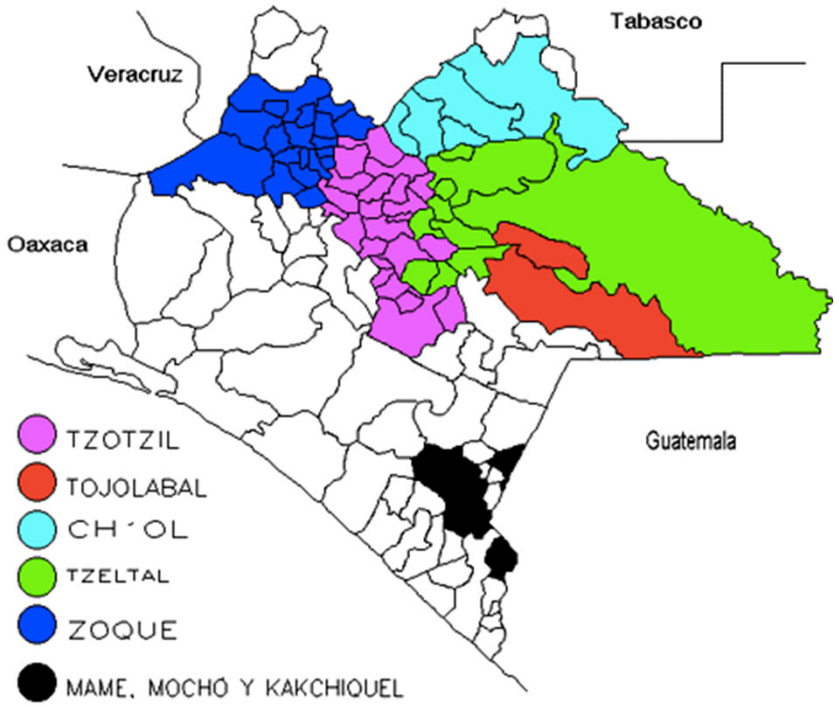


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

Thinking about space…



It seems natural in most languages to talk about space egocentrically when talking about things that are nearby.

What is to your left or to your right, in front of you or behind you, etc.

Not so in Tzeltal, a Western Mayan language spoken in Chiapas.

In this language (and a small number of others, mostly unrelated to one another, and broadly distributed, e.g., in Bali and Queensland) absolute directions are used.

Raise your east hand if this seems intuitive to you…



Some native Tzeltal speakers defending their land from the vampire squid of neoliberalism.



Beautiful downtown Tenejapa.

Tzeltal has a historically landmark-based system.

Things are either “uphill” or “downhill” from one another, not to the left or right of the speaker or interlocutor.

In fact, “uphill” and “downhill” are used as cardinal directions, making this an *absolute* system.

At night, in an alien city, facing a device never seen before (namely a sink with two taps), one Tenejapan asked another, “Which is the hot tap, the uphill (southern) or the downhill (northern) one?”. They maintain a constant sense of absolute orientation, presumably by running a continuous background computation of egocentric heading with respect to abstract bearings, integrating multiple internal and external cues to achieve this.[17] This is the phenomenon that we are trying to capture.

(Levinson et al., 2002, Cognition)

A non-linguistic test for relative vs. absolute coding of space

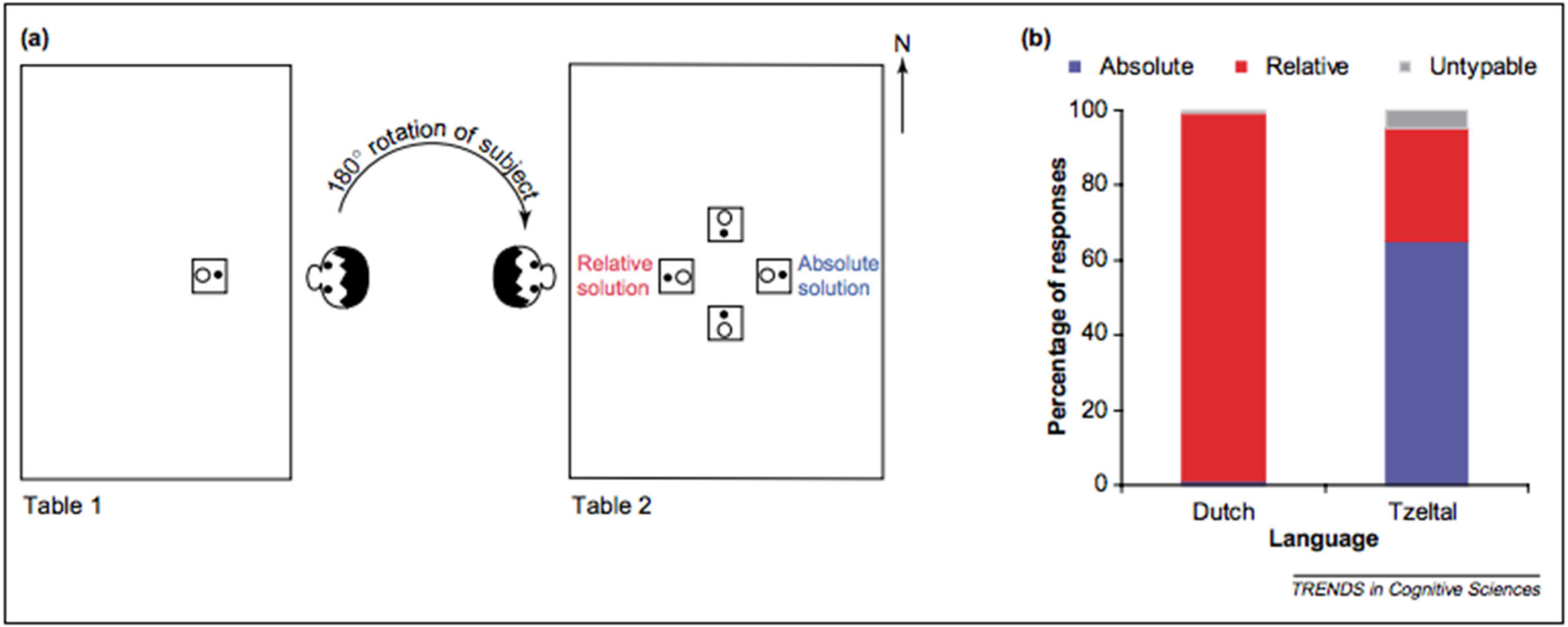


Figure 2. Memory for spatial configuration: 'the chips task'. **(a)** Participants saw on Table 1 a card printed with a large and a small dot arranged with the small dot towards them, away from them, to the left, or to the right. After a 30-s delay, they were rotated through 180° and led to Table 2, where they were asked (in the local language) to identify from a set of four cards the similar or counterpart card to the one they had seen before. Each participant had eight trials, which varied in the arrangement of the dots. Responses that preserved sameness in egocentric coordinates were coded Relative, those that preserved Absolute coordinates were coded Absolute, and other responses were coded 'Untypable'. **(b)** The results for just two languages, Dutch and Tzeltal. The Relative/Absolute trend matches the preferred linguistic FoR: Dutch responses were overwhelmingly Relative whereas Tzeltal responses were overwhelmingly Absolute. The proportionately larger Tzeltal inconsistency can be attributed to the fact that the data are from an unschooled peasant population. However, if we look for 'consistent coders', that is, coders who give the same response on 6 out of 8 trials, then over 80% of Tzeltal speakers are 'Absolute thinkers' (Adapted from [13], pp. 159–160, by permission of Cambridge University Press).

Majid et al. (2004, TiCS)

You can get really fancy with this.

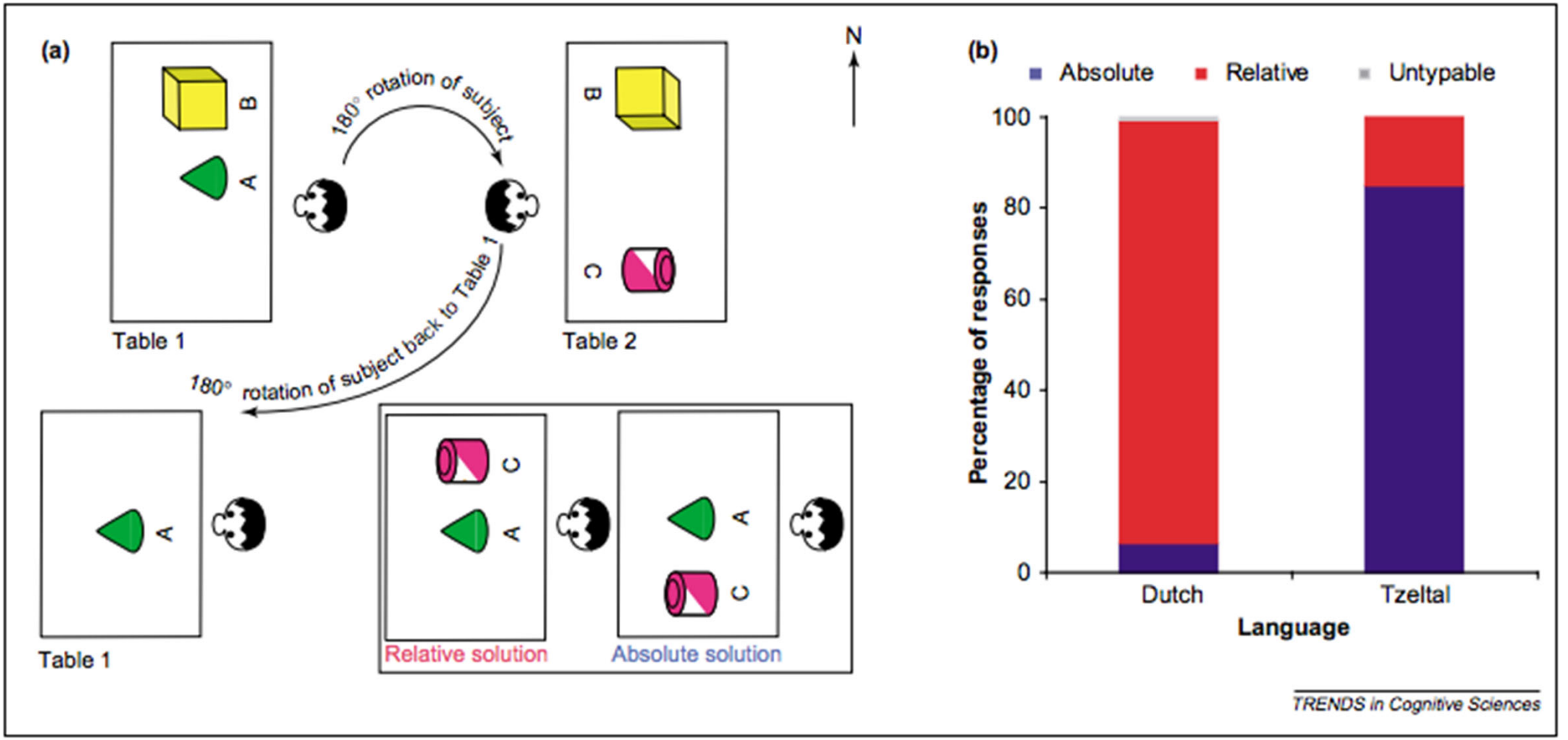


Figure 4. Spatial reasoning: making a transitive inference. **(a)** Transitive inferences (e.g. if A is bigger than B, and B is bigger than C, then A is bigger than C) can be drawn on the basis of non-linguistic arrays. In this experiment, participants saw a cube and a cone arranged in a particular configuration on Table 1, and were then rotated through 180° to Table 2 where they saw a cube and a cylinder. Finally they were rotated back to Table 1, where the cone was standing alone. Their task was to place the cylinder next to the cone, keeping the location consistent with what they had seen before. There were five trials using the transverse axis. There are two ways to perform this task: one using Relative coding (the cube is to the right of the cone, and the cylinder is to the right of the cube, therefore the cylinder is to the right of the cone), and the other using Absolute coding (the cube is to the south of the cone, and the cylinder is further south of the cube, therefore the cylinder is to the south of the cone). **(b)** Once again, the results for Dutch and Tzeltal show that the Relative/Absolute trend matches the preferred linguistic FoR: Dutch participants gave Relative responses whereas Tzeltal participants gave Absolute responses (Adapted from [13], pp. 162–167, by permission of Cambridge University Press).

Majid et al. (2004, TiCS)

Li and Gleitman:
Inducing landmark-based similarity in English speakers

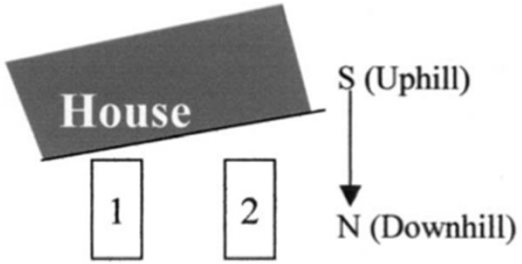


Fig. 4. Test environment for the Tenejapan subjects (following Brown & Levinson, 1993).

Consider the conditions under which the Tzeltal speakers were tested:
Outside, with visible, familiar landmarks.

Typical psychology lab studies are done in windowless rooms.

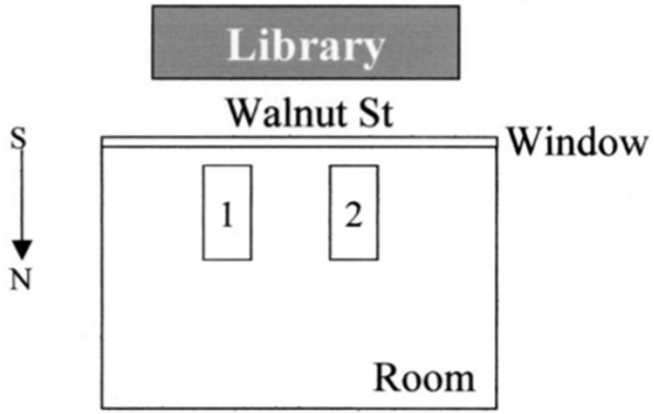


Fig. 5. Test environment for the indoor conditions of Experiment 2a. The University Library is to the south of the tables just as was the house in Fig. 4.

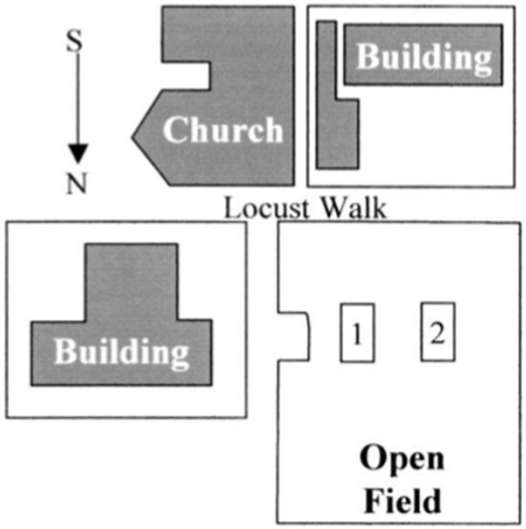


Fig. 6. Test environment for the Outdoor condition of Experiment 2a. Buildings are to the south and southeast of the grassy test area.

Li and Gleitman:

Inducing landmark-based spatial reasoning in English speakers

In the “blinds down” condition, English speakers overwhelmingly use relative position in this task – like the Dutch speakers in prior studies.

With the blinds up and a view of the street, or outdoors, their preference is less pronounced.

In addition to recording subjects' spatial solution to the problem, we also noted whether they asked for clarification during the test or mentioned it during a post-interview. We found that in the Blinds-Down condition (closest to Brown and Levinson's Dutch study), only in 20% of cases did subjects ask for clarification during training ("Should I put them still going the same way, or still going to my left?"). In the other, landmark-rich, conditions, the ambiguity of the task leapt out at the subjects not only in their response styles (the bimodal result shown in Fig. 7) but also in the 70% instance of spontaneous queries during the training phase. Though explicit querying seems to have been rare in the traditional populations, it did sometimes occur (Wilkins, e-mail communication, June 13, 2000; Levy, e-mail communication, November 27, 2000).

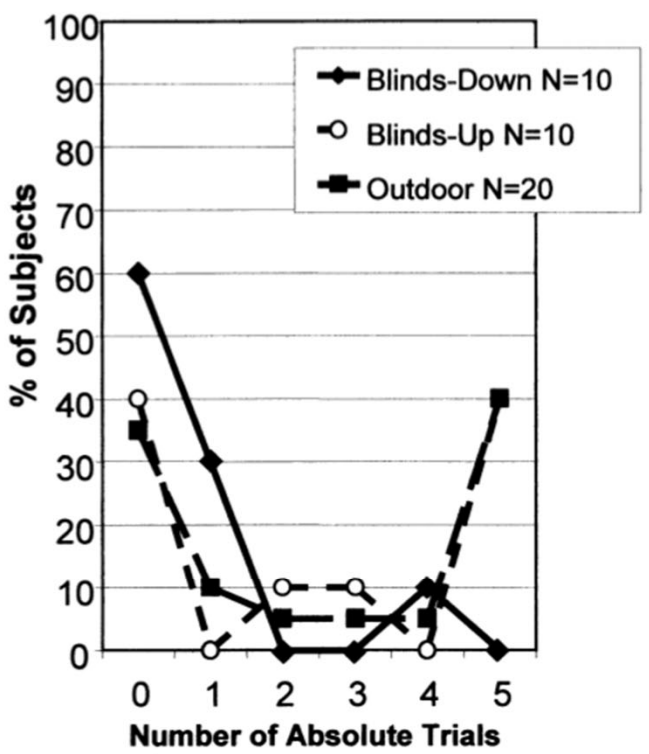


Fig. 7. Proportions of absolute choices in Blinds-Down, Blinds-Up, and Outdoor conditions. Subjects in the Blinds-Down condition predominantly chose the relative response. In the other conditions, subjects were divided in their preference for either all absolute responses or all relative responses.

Of all of our experiments, Li and Gleitman have chosen to replicate the very simplest ('Animals-in-a-row' – see Levinson (in press) for the many converging results from other experiments) and have gone on to simplify it further. The task in essence consists of presenting participants with a row of three animals on a table, rotating the participants 180 degrees, and making them reconstruct the array on another table so that to their satisfaction it matches the first. In our experiment, it was a crucial part of the design that participants' attention was deflected from the *direction* of the stimulus by being required to memorize the *order and identity* of three toy animals drawn from a larger set of four (Levinson, 1996b, p. 114). It was presented as a memory test, first without rotation, then with rotation (and both accuracy of order and direction were coded), and the participant was walked up to 20 m between stimulus and response.

The details of the Li and Gleitman replication vary from the original, including no translation of the participant and no delay after removal of the stimulus and thus a considerably shorter period for retention in memory,[8] and most importantly, the participant's memory task was reduced by presenting the subject with the three animals used in the stimulus, not the full set of four.[9] As we shall show, memory load can make a major difference (see our Experiment 2 below), and in addition when the experiment becomes too transparent participants may second-guess the experimenter's interest in direction rather than order or type of animal. It seems clear from the Li and Gleitman report that many of their participants were second-guessing the experimenter's intentions – they queried the instructions in a way that suggests that direction was clearly at issue. It is always hard to design a task that is matched across schooled and unschooled subject populations, but this should have been a warning that the task was not sufficiently opaque, for what we are interested in is participants' non-reflective utilization of a spatial coding scheme for memory, not what they think the experimenters think they should be doing. Incidentally, for this reason in our original Dutch experiments we used 40 participants of mixed ages (21–77), sex and occupational background, like the participants in our cross-cultural studies.

Levinson and colleagues were not impressed.

They claim that Li and Gleitman made it too transparent that they were testing something to do with rotation by reducing memory load and time/distance between stimulus and test.

Also Li and Gleitman's participants were Ivy League students, who excel at figuring out what the professor wants and then making it happen.

Research question: Is there categorical perception for these dimensions?

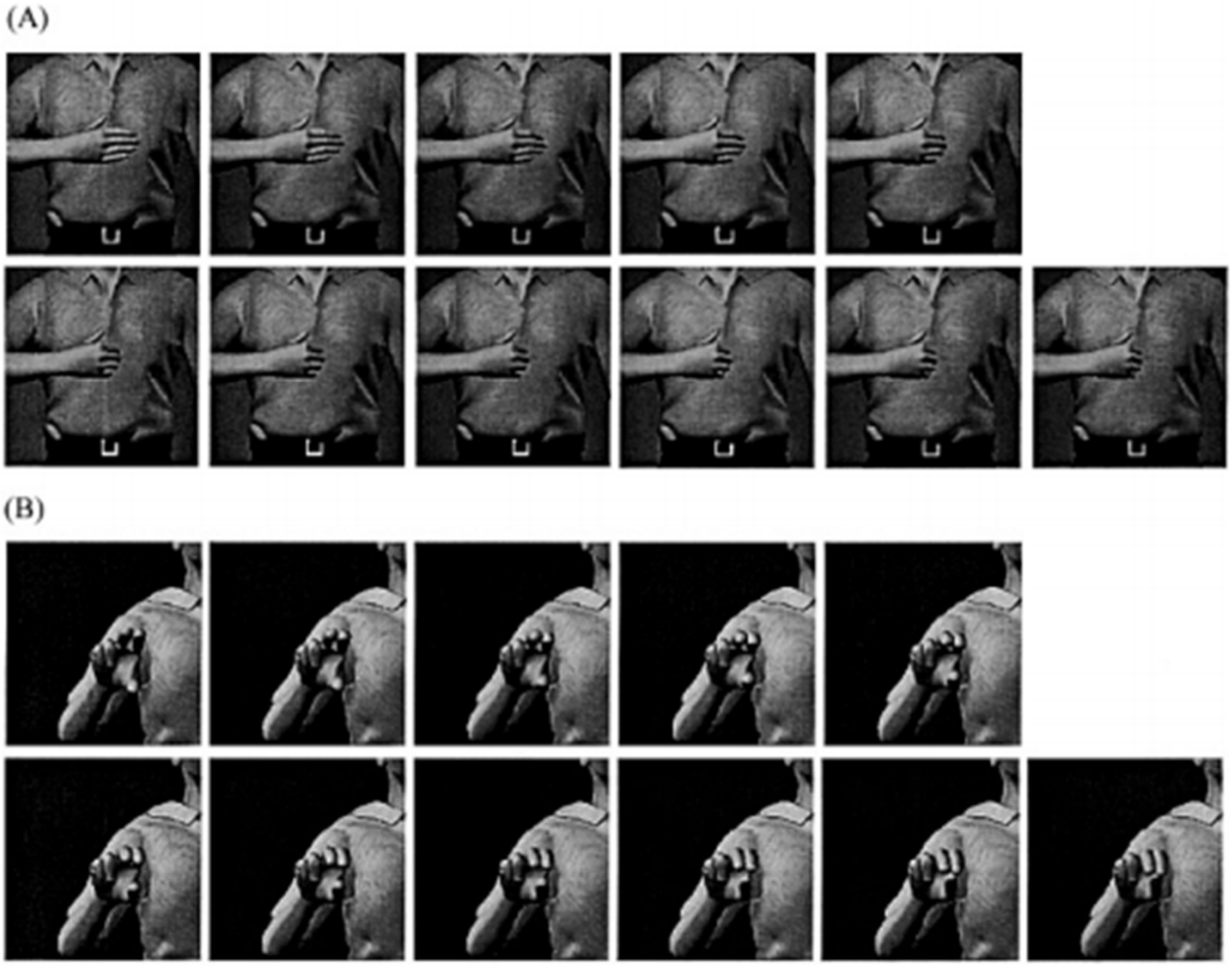
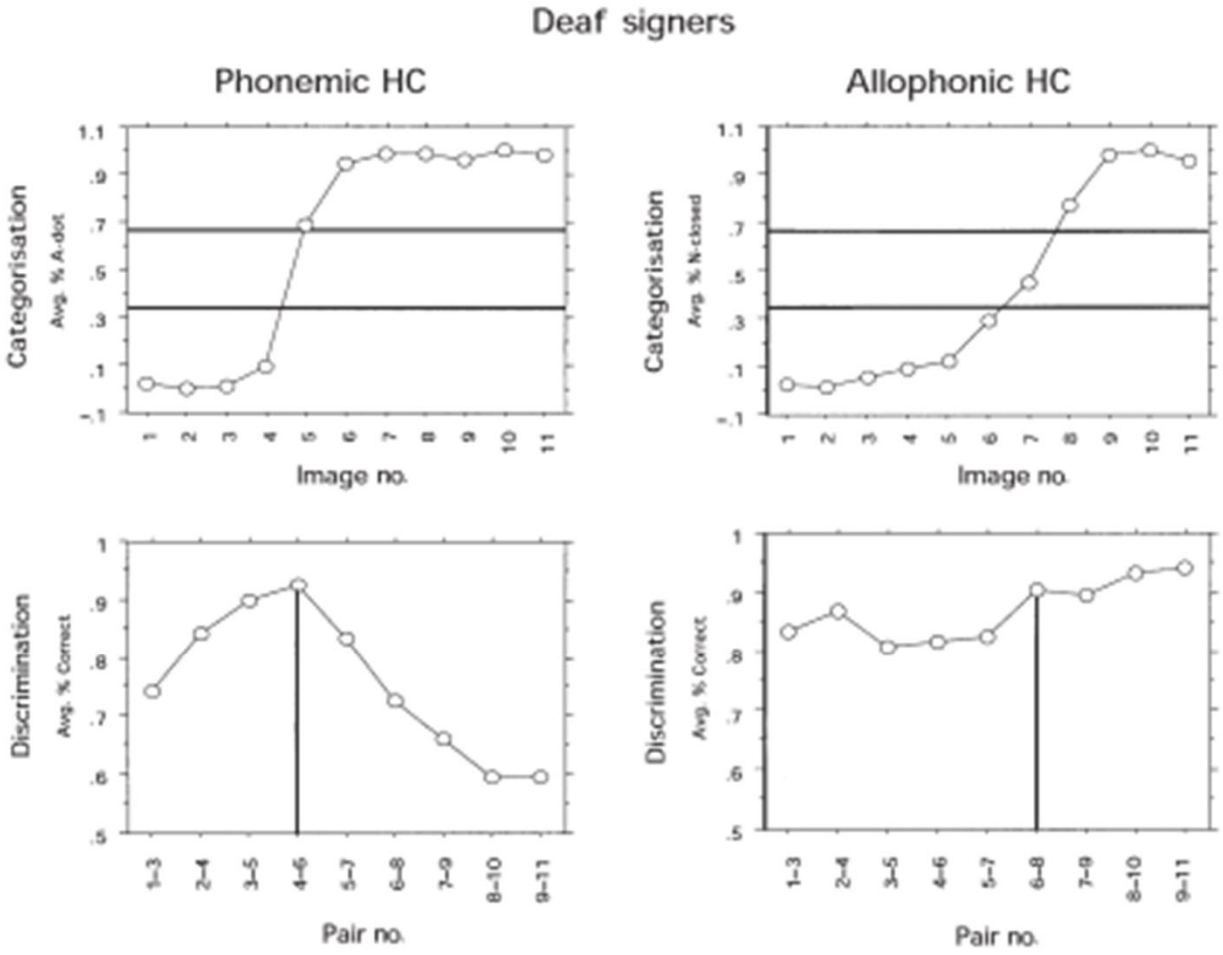


Figure 1. Illustration of stimulus continua varying in (A) phonemic hand configuration (from the sign PLEASE to the sign SORRY) and (B) allophonic hand configuration (from 'open-N' to 'closed N' in the sign SAY-NO-TO). The continua were created with Poser software from MetaCreations.



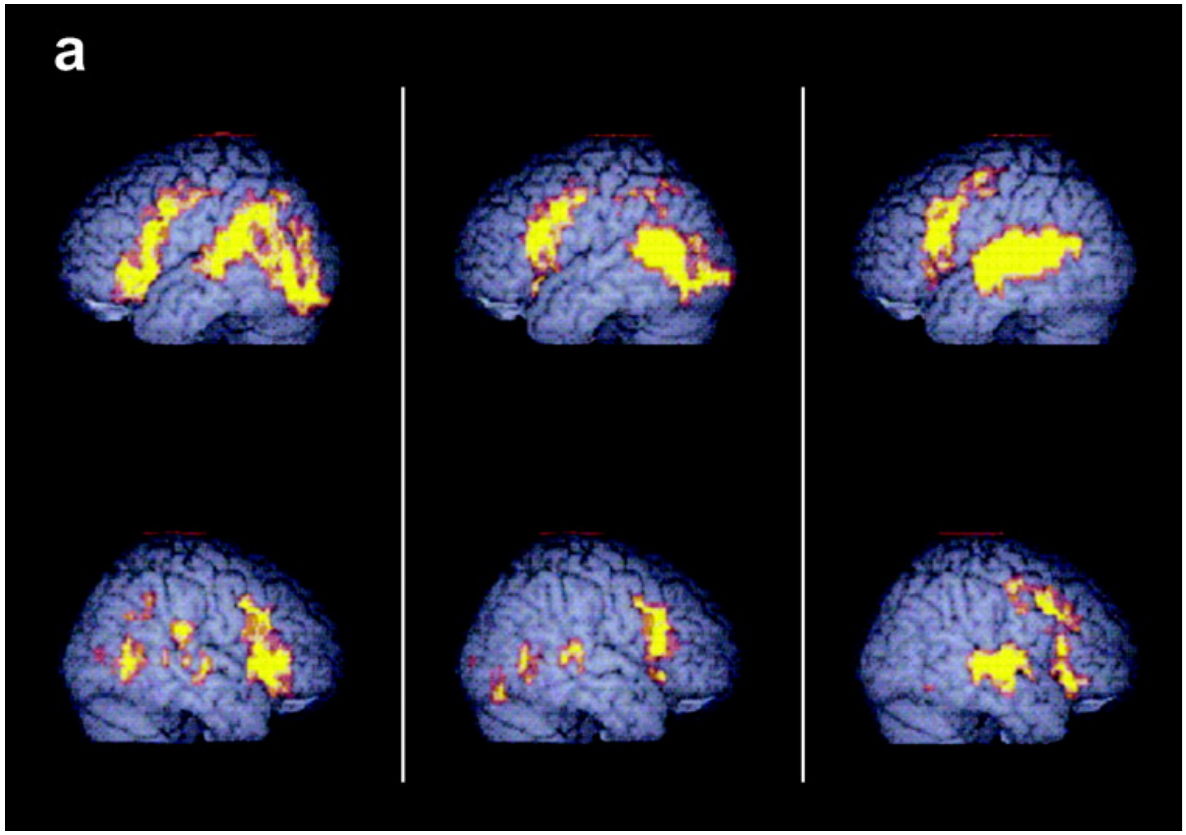
These curves should look familiar.

There is a fairly sharp categorical boundary between stimuli 4 and 6 for the phonemic boundary.

This corresponds to the peak in the discrimination function.

For the allophonic condition, the identification function is much shallower – i.e., many more ambiguous stimuli – and there is no peak in the discrimination function.

Brain activity during sign language comprehension has both important similarities and differences with spoken language.



Deaf signers

Hearing signers

Audio-visual
Speech (hearing participants)

Similar:

- left lateralization
- classical language areas

(Broca's / inferior frontal; Wernicke's / posterior superior temporal)

Different:

- +visual areas (-auditory areas)
- superior parietal / classic spatial areas

Campbell, MacSweeney, & Waters (2008)

So, sign language…

Sign language has phonological structure, and native speakers show categorical perception for some contrastive features.

Learning a signed language results in some brain reorganization, particularly the lateralization of motion processing.

Although attention to the visual periphery is critical for both hearing and deaf signers, only deaf signers show enhanced processing of visual attention in the periphery.

Compared with non-language stimuli that also involve organized, sequenced hand movements, signed sentences activate posterior STG (Wernicke's area).

Two versions of the “critical period” hypothesis

Version One: The exercise hypothesis. Early in life, humans have a superior capacity for acquiring languages. If the capacity is not exercised during this time, it will disappear or decline with maturation. If the capacity is exercised, however, further language learning abilities will remain intact throughout life.

Version Two: The maturational state hypothesis. Early in life, humans have a superior capacity for acquiring languages. This capacity disappears or declines with maturation.

Which of these do you think JN89 see as more consistent with their data?

Does this change our interpretation of the L1 critical period data?

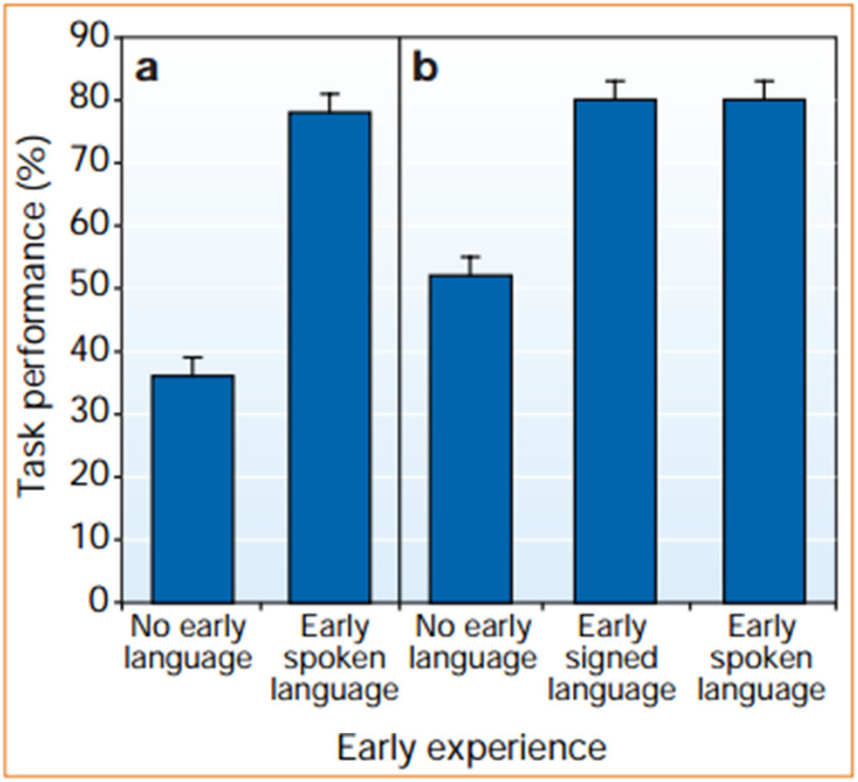


Figure 1 Effects of early experience on later language learning. **a,** American Sign Language (ASL) performance of deaf adults who had experienced no language in early life, and of deaf adults who had experienced spoken language in early life. Subjects were tested using a task requiring recall of complex ASL sentences. **b,** English performance of deaf adults who had had no experience of language in early life, of deaf adults who had experienced ASL in infancy, and of hearing adults who had experienced a spoken language other than English in infancy. Subjects were tested using a task requiring judgements of whether complex English sentences given in print were grammatically correct; chance performance is 50%. Further details are available from the authors.

Many deaf children do not get sign language input, but have an otherwise typical upbringing.

Children (ages 9-15) have difficulty learning American Sign Language (ASL) if this is their first time learning a language.

A comparison group of children who went deaf after initially learning a spoken language do much better.

These differences persist into adulthood.

Mayberry, Lock, & Kazmi (2002, Nature)

A criticism sometimes raised against the semantic blocking paradigm as a means of studying lexical-semantic processes is that the blocking effects may arise earlier, during the visual-conceptual analysis of the items, rather than during lexical processing. One might argue that it is perhaps more difficult to discriminate between items belonging to the same semantic category, which are likely to share visual features, than between items belonging to different semantic categories. As noted above, the items used in the present experiment were selected to be as dissimilar as possible. For 12 of the 16 items we used, Damian et al. (2001) collected similarity ratings, which revealed that the items in homogeneous sets were, on average, not judged to be more visually similar than those in heterogeneous sets. In addition, the absence of a blocking effect during the first presentation of the materials argues against the allocation of the blocking effect at the visual-conceptual level: It is difficult to see why an effect of visual similarity would be absent at first and arise after the first presentation of the materials. Instead, one might expect to see the opposite pattern—visual similarity among items should be more influential at the beginning of a block, when the participants are uncertain which features discriminate between the objects, than after they have become more familiar with the objects. For a supplementary experiment with eight participants we selected items that were semantically as well as visually highly similar objects (e.g., shirt, blouse, coat, jumper; see Appendix B). We only included four presentation cycles because Experiment 1 of the present study had shown that blocking effects emerge after the first presentation cycle and remain stable thereafter. Using the visually similar items we found a context effect in the first as well as all following repetitions, resulting in significant main effects of context, $F(1, 7) = 15.24$, $MSE = 3{,}335$, $p < .01$, and presentation cycle, $F(3, 21) = 24.93$, $MSE = 2{,}075$, $p < .001$, but no significant interaction of context and presentation cycle. These findings support our assumption that semantically and visually similar sets should produce a different pattern of blocking effects across presentation cycles than should semantically similar, visually dissimilar sets.

Dealing with potential confounds 101

What if the blocking effect arises at the level of visual object recognition?

Two responses:

1. Stimuli were normed so that objects in the same category were not more visually similar to one another than objects in different categories, on average.

2. A control experiment was run in which visual similarity was blocked along, and *different results were found.*

Specifically, the blocking effect was observed in *the first block of trials*.

Thus, the effects in Experiment 1 aren't likely to be due to visual similarity.

Yeah, but what about that Hermens et al., study, where people saw multiple objects simultaneously, and they didn't find an effect?

Maybe there's just something weird about seeing objects in serial?



Belke et al. address this by asking participants to view displays of four items simultaneously and produce their names in serial.

They measure eye-movements while they do this.

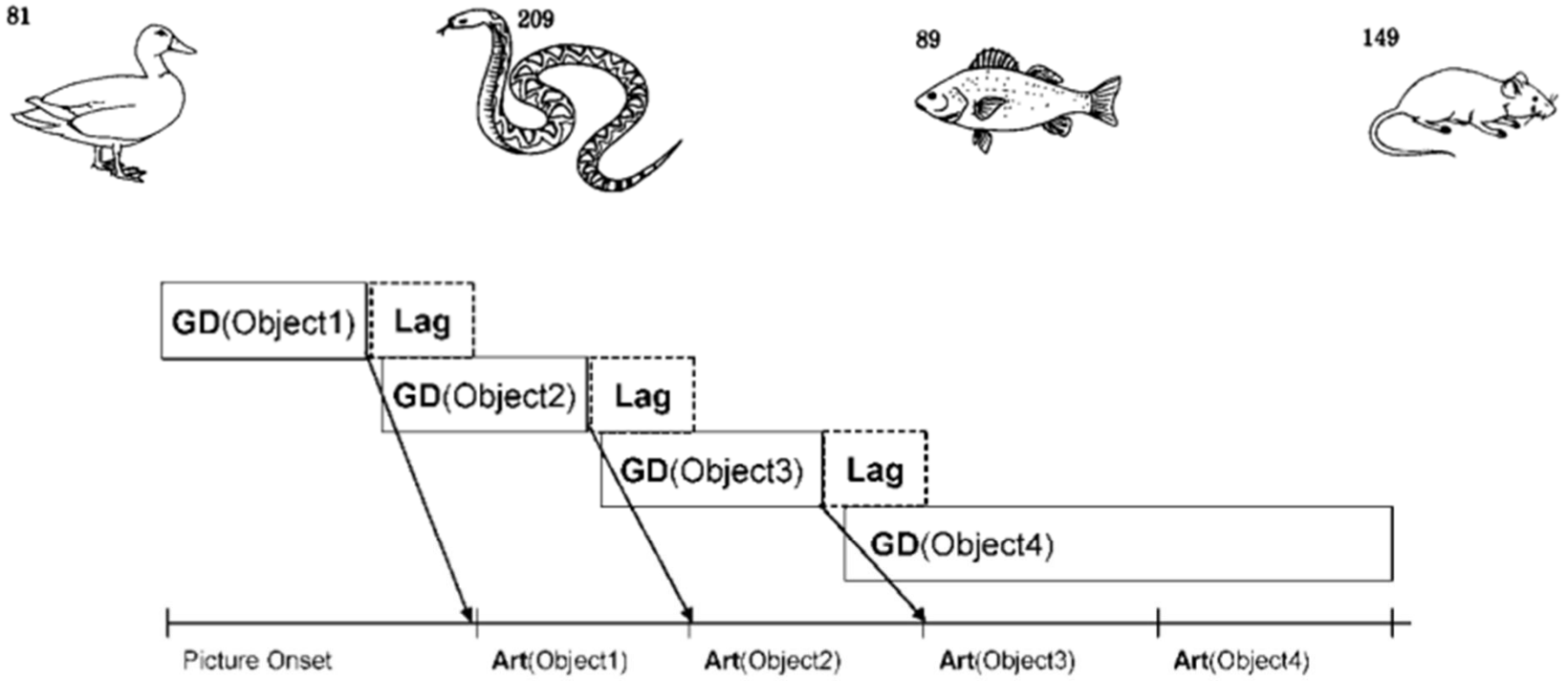


Figure 3. Schematic overview of the temporal coordination of gaze durations (GD) and articulatory word onsets (Art) in a multiple-object naming task.

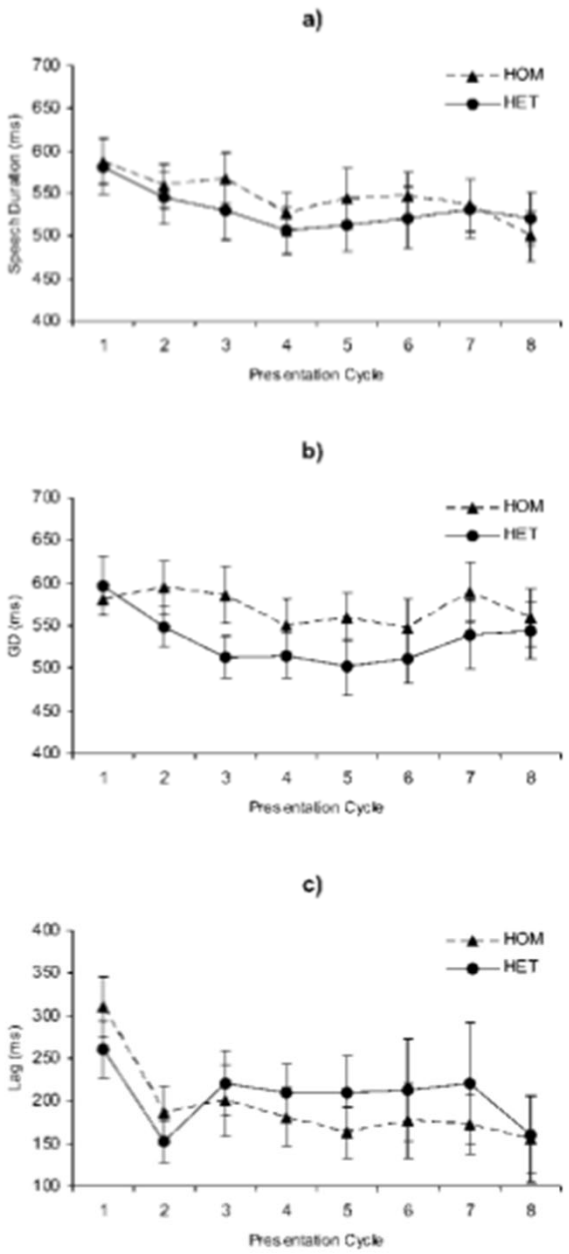


Figure 4. Experiment 2: (a) Mean speech durations, (b) gaze durations (GD), and (c) eye–speech lags by context and presentation cycle. The error bars represent one standard error.

Overall duration of speech decreases over the course of the experiment, but doesn't show a blocking effect.
→*gesture level*

Gaze duration shows a blocking effect, but only after the first cycle.
→*lemma selection*

There's no effect on lag. Numerically, the blocking effect seems to go in the opposite direction, but this is not significant.
→*phonological encoding*

So, we want to know, is phoneme identification for word recognition more like thresholded (lumber mill) or cascaded (food web) processing?

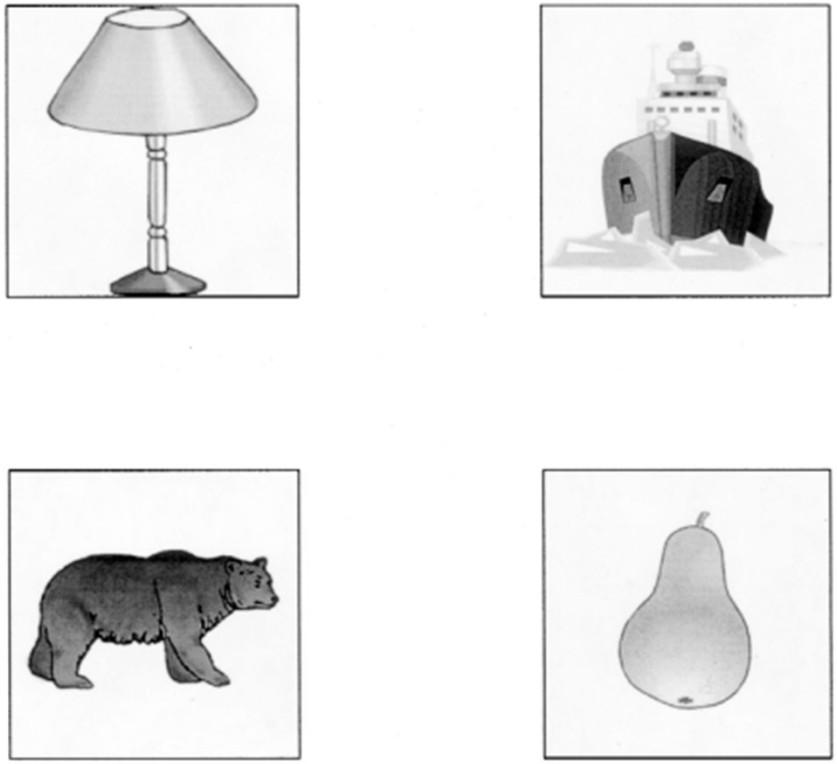


Fig. 3. A sample display. Each display contains a target (e.g., the "pear"), a competitor (e.g., the "bear"), and two filler items ("lamp" and "ship").

McMurray et al. tested identification of a /b/-/p/ continuum in a "visual world" display for eye-tracking.

McMurray, B., Tanenhaus, M., Aslin, R. and Spivey, M. (2003). Probabilistic constraint satisfaction at the lexical/phonetic interface: Evidence for gradient effects of within-category VOT on lexical access. *Journal of Psycholinguistic Research, 32*, 77-97.

http://www2.psychology.uiowa.edu/faculty/mcmurray/publications/mcmurray-jpr.pdf

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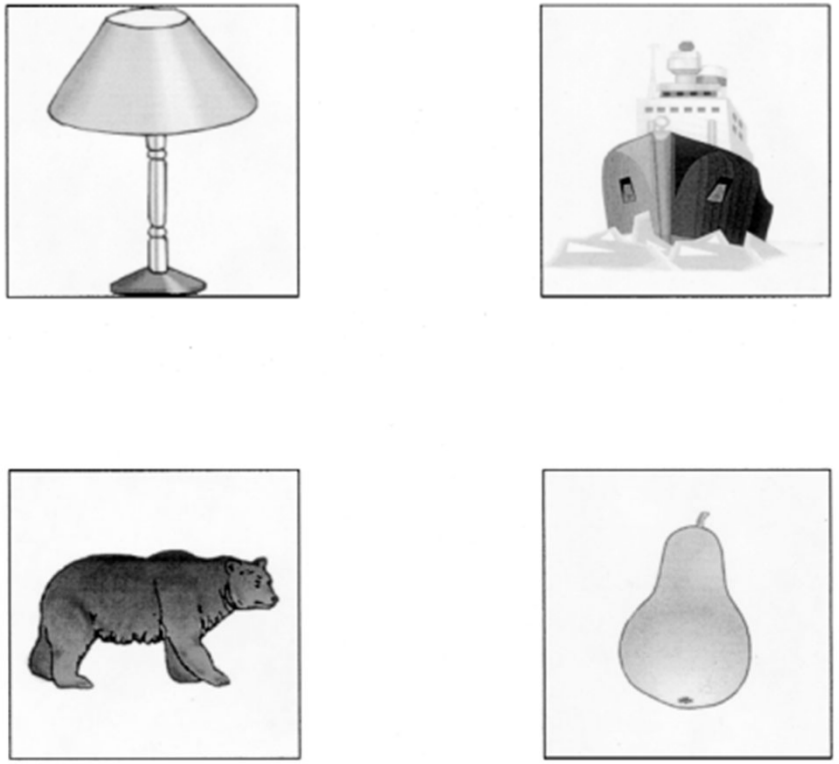


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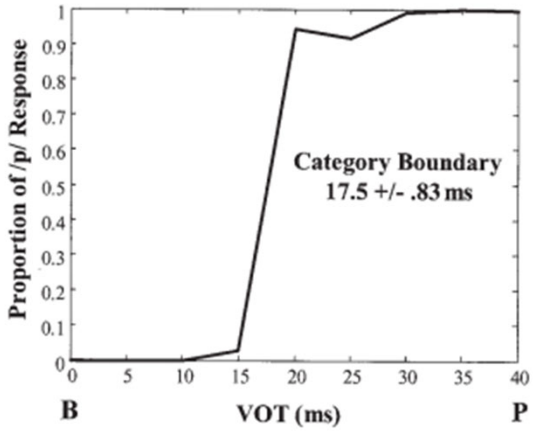


Fig. 1. Identification function (from mouse responses) as a function of VOT.

Participants used the mouse to select the picture most consistent with what they heard, e.g., a "bear" or a "pear," just like in the experiment you did at home.

A typical, step-like, identification function was observed.

Eye-movements reveal that ambiguity is *gradient,* i.e., there are more looks to the competitor close to the boundary.

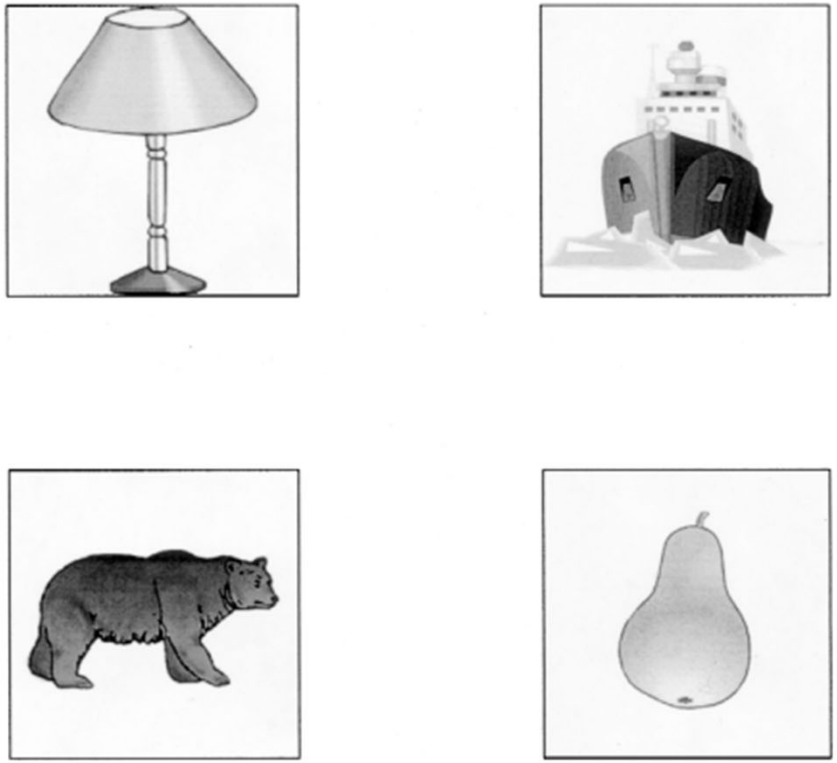


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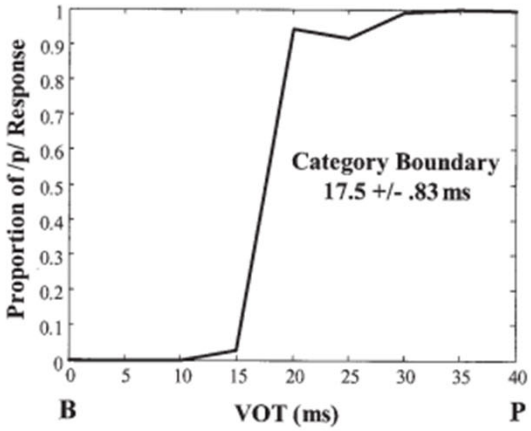


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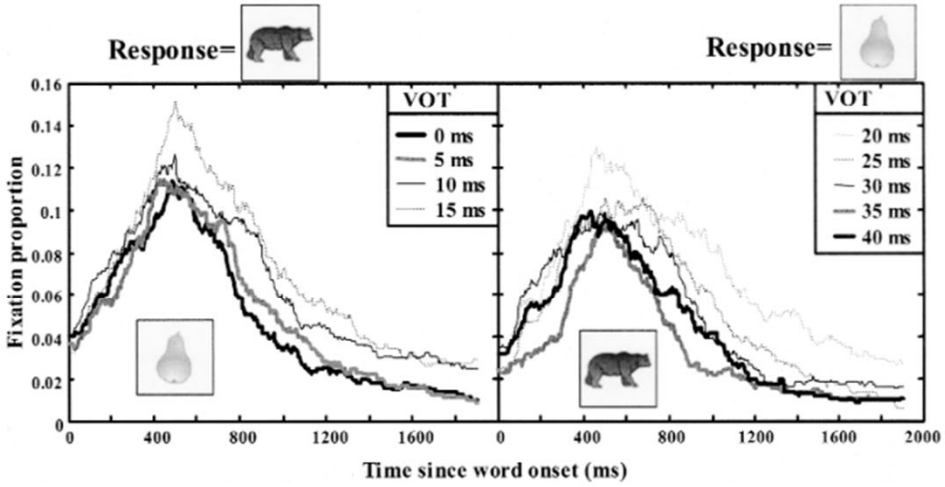


Fig. 7. Effect of VOT and time on fixations to the competitor. A clear gradient effect of VOT can be seen. Importantly, the effect of VOT is primarily on the duration of activation.

The effect is a lot clearer viewed this way…

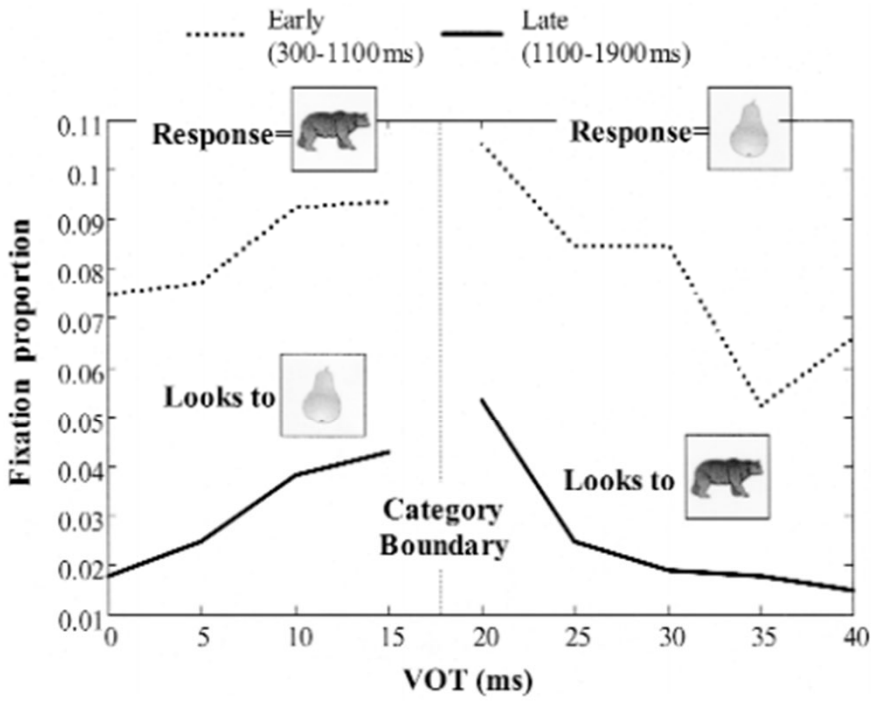


Fig. 6. Effect of VOT and time on fixations to the competitor.